

Entrepreneurial Heterogeneity, Financial Development, and Business Cycle Dynamics

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First Version: May 21, 2024

This Version: Aug 3, 2026

Abstract. This paper studies how financial development affects business cycles through entrepreneurial composition. Using Argentine microdata, I show that self-employed entrepreneurs are concentrated below the median of the income distribution and have countercyclical population shares, while employer entrepreneurs are concentrated above the median and have procyclical shares. Motivated by these patterns, I develop a dynamic occupational choice model with aggregate risk in which endogenous sorting across occupations shapes macroeconomic fluctuations. In the calibrated model, financial development raises aggregate output volatility by 8 percent because relaxing collateral constraints shifts economic activity toward employer entrepreneurs, who are more sensitive to aggregate shocks. These findings show that while financial development raises long-run output, it may also increase short-run instability.

Keywords: Entrepreneurs; Heterogeneous agents; financial development

JEL Classification: E32, F41

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1 Introduction

Entrepreneurship plays a vital role in economic development, yet these activities are far from homogeneous, with a diverse range of entrepreneurs operating at different scales and facing unique challenges. While a large body of literature has studied the long-run effects of financial development policies on aggregate output and productivity, its impact on short-run macroeconomic stability is less understood. This paper addresses this gap by investigating the business cycle consequences of financial development in an economy with entrepreneurial heterogeneity.

I show that financial development can generate a trade-off between long-run outcomes and short-run stability. While improved access to credit enhances the steady-state of the economy, it simultaneously amplifies business cycle volatility through an endogenous compositional shift in entrepreneurial activity. Specifically, financial deepening encourages a reallocation from self-employed subsistence entrepreneurs toward employer entrepreneurs who operate larger firms and respond more aggressively to aggregate shocks. This compositional change increases the economy's sensitivity to business cycle fluctuations, creating larger volatility in the aggregate.

The analysis is motivated by two empirical facts documented using microdata from Argentina's Household Survey (EPH). First, entrepreneurial composition varies systematically across the income distribution: self-employed entrepreneurs are disproportionately concentrated below the median, while the majority of employer entrepreneurs reside above the median. Second, the aggregate share of entrepreneurs exhibits strong countercyclical behavior, driven primarily by the countercyclical population share of self-employed entrepreneurs. These patterns suggest that response of entrepreneurs to business cycles are heterogeneous across types.

To formalize these insights, I develop a dynamic general equilibrium model with occupational choice, entrepreneurial heterogeneity, and aggregate risk. The framework incorporates three key mechanisms that generate realistic patterns of entrepreneurial sorting and cyclical behavior. First, labor market frictions create barriers to employment transitions, giving rise to a class of necessity-driven entrepreneurs, primarily self-employed,

who use entrepreneurship as insurance against unemployment risk. Second, I distinguish between self-employed and employer entrepreneurs through endogenous labor hiring decisions and fixed operating costs, which naturally sort individuals by wealth and productivity. Third, correlated aggregate productivity and financial shocks generate business cycle fluctuations that affect both production technologies and credit market conditions.

The model is calibrated to match key features of the Argentinian economy, including cross-sectional income inequality and the distribution of individuals across occupational categories. The quantitative framework successfully replicates the cyclicity of population shares and aggregate income dynamics across occupational groups observed in the microdata. Importantly, the model generates endogenous heterogeneity in shock transmission: employer entrepreneurs exhibit larger responses to both productivity and financial shocks compared to the self-employed group.

Next, I conduct a counterfactual experiment that models financial development as a loosening of collateral constraints. This setup captures a pure credit expansion, holding the volatility of aggregate shocks fixed.² Because the model allows occupational choices and equilibrium prices (wages and interest rates) to adjust endogenously, it captures general equilibrium feedbacks that a purely reduced-form empirical exercise cannot recover. The quantitative results show that financial development raises long-run output but also increases aggregate volatility. For example, an increase in the external credit-to-GDP ratio from 0 to 2 raises output volatility by more than 8 percent. The mechanism is driven by a compositional shift in the exposure to the business cycle: relaxing collateral constraints reallocates output, capital, and labor toward employer entrepreneurs, who operate at larger scale, hire external labor, and respond more strongly to aggregate shocks. Although financial development dampens output volatility within the employer sector, employers remain the most volatile sector throughout the experiment. As a result, the rising aggregate weight of employer entrepreneurs dominates this sectoral stabilization and increases aggregate volatility. Overall, the analysis highlights that financial development raises long-run output while reallocating aggregate risk towards larger, more shock-sensitive

²In practice, however, episodes of financial deepening often coincide with regulatory and institutional reforms that directly affect the volatility of shocks themselves. Nonetheless, this exercise provides a useful and stylized counterfactual.

entrepreneurs.

This paper builds upon and contributes to several strands of literature at the intersection of entrepreneurship, financial frictions, and business cycles. First, this paper draws from the literature on entrepreneurship and occupational choice. This research has established that financial constraints significantly impact entrepreneurial entry and outcomes (Evans and Jovanovic (1989) and Quadrini (2000)) and has explored how individual characteristics shape the choice to become an entrepreneur versus a wage worker (Lucas (1978)), with important consequences for wealth and income inequality (Cagetti and De Nardi (2006)).

Second, I contribute to the literature studying the aggregate consequences of misallocation and financial frictions. A key theme in this work is that improving allocative efficiency through financial development can lead to substantial gains in productivity and output (Midrigan and Xu (2014) and Hsieh and Klenow (2009)). Consequently, a large body of research has analyzed the steady-state effects of policies that ease financial constraints, such as broad financial reforms as in Buera et al. (2011) or microfinance programs as in Buera et al. (2021) and Herreño and Ocampo (2023). This paper extends this analysis by focusing on the business cycle implications of such policies, moving beyond a purely steady state perspective.

Third, the paper is closely related to an active literature on subsistence entrepreneurship, informality, and labor market frictions in developing economies. This work emphasizes that many entrepreneurs are driven by necessity rather than opportunity. Several studies, such as Poschke (2012) and Feng and Ren (2023), have explored how factors like education or labor market frictions, such as Donovan et al. (2023), Finamor (2025), and Herreño and Ocampo (2023), give rise to a large class of low-income, self-employed individuals. These “necessity” entrepreneurs often act as a substitute for formal unemployment insurance as noted by Jaar (2023). Papers such as McKiernan (2021) study how the presence of informal sectors interact with public policies in complex ways. Furthermore, Bosch and Maloney (2010) and Loayza and Rigolini (2011) have documented countercyclicality of self-employment in emerging markets. While some structural models have explored the business cycle implications of informality (see Finkelstein–Shapiro (2014),

Restrepo-Echavarria (2014), Horvath (2018), and Fernández and Meza (2015)), these have typically been in representative-agent frameworks. My contribution is to analyze these dynamics in a setting with uninsurable idiosyncratic risk and rich household heterogeneity.

Perhaps most closely, this paper extends the frameworks of Herreño and Ocampo (2023) and Allub and Erosa (2019). I build on the former by incorporating labor market frictions to “push” poor households into entrepreneurship. I build on the latter by differentiating between self-employed and employer entrepreneurs to generate an endogenous sorting of rich and poor entrepreneurs. My primary contribution is to integrate these two frameworks and introduce aggregate risk to study the business cycle dynamics that were outside the scope of those papers. While Kwark and Ma (2020) also studies entrepreneurship with aggregate risk, my model features a household problem that includes these crucial labor market frictions and entrepreneurial heterogeneity.

Finally, the paper contributes methodologically by applying state-of-the-art computational techniques for solving heterogeneous-agent models with aggregate shocks. Following the approach pioneered by Reiter (2009) and further extended by Bayer and Luetticke (2020), I use a combination of non-linear methods for the stationary equilibrium and linearize the model only with respect to aggregate shocks. To the best of my knowledge, this is the first paper to apply the Reiter’s method to a model of occupational choice to study the business cycle impacts of financial development.

The outline of the paper is as follows: Section 2 presents the empirical evidence on entrepreneur heterogeneity. Section 3 presents a heterogeneous-agent model with occupational choice and labor market frictions with aggregate shocks. Section 4 presents the solution method used to solve the model and the calibration strategy. Section 5 discusses quantitative results. Section 6 concludes.

2 Empirical Evidence on Entrepreneur Heterogeneity

In this section, I provide the empirical motivation for the model by documenting two key features of entrepreneurship using an Argentine dataset. First, I show that the compo-

sition of entrepreneurs differs systematically across the income distribution, with shares of self-employed entrepreneurs declining as income rises. This pattern is consistent with Herreño and Ocampo (2023), who document a U-shaped relationship between entrepreneurship and income in economies with large share of self-employment. Second, consistent with a broad literature (e.g., Bosch and Maloney (2010); Loayza and Rigolini (2011)), I find that the aggregate share of entrepreneurs is countercyclical. This dynamic is driven primarily by self-employed entrepreneurs, who constitute the majority of entrepreneurs and have lower average incomes than their employer counterparts. Together, these facts underscore the importance of distinguishing between self-employed and employer entrepreneurs to capture key distributional patterns and aggregate dynamics.

Data Description and Summary Statistics

The microdata for this analysis is from the Encuesta Permanente de Hogares (Permanent Household Survey, or EPH), conducted by the National Institute of Statistics and Censuses (INDEC) of Argentina. I selected this dataset for two reasons. First, Argentina has a significant prevalence of self-employed entrepreneurs, the focal point of this study. Second, the EPH has been available quarterly since 2004, providing a rich, high-frequency dataset that is ideal for business cycle analysis, in contrast to the annual data common in other individual-level surveys.³

This study focuses on the period from 2004Q1 to 2015Q2 to account for methodological revisions to the EPH post-2015. Data for 2007Q3 is absent due to an EPH staff strike; where necessary, I linearly interpolate this data point. The survey provides comprehensive data on individual characteristics related to employment, occupation, and income. This analysis relies on a few key variables to classify individuals. The variable, *CAT_OCUP*, identifies a person’s primary occupation as an employee, a self-employed entrepreneur, or an employer entrepreneur. To identify unemployed individuals, I use the variable *PP10A*, which records how long an individual has been looking for a job.

³While the EPH data extends back to 1998, available biannually until 2003, this study excludes the pre-2004 data due to inconsistencies in survey frequency. The main results are robust to including the pre-2004 data.

Total income is measured using variable *P47T*. This variable should be interpreted as reported monthly individual income: it includes income from the respondent's main occupation as well as other labor and non-labor income sources. Appendix A.1 and A.2 provide additional details on variable construction and related measurement issues.

Finally, data on aggregate variables such as GDP and consumption come from the International Monetary Fund's (IMF) International Financial Statistics (IFS) database. All aggregate series are logged and linearly filtered. The results are robust to alternative filtering methods.

Table 1: **Summary Statistics**

	Unemployed	Workers	Self-employed	Employer
Income Share	0.01	0.77	0.15	0.07
Population Share	0.10	0.69	0.17	0.04
Income Share/Population Share	0.10	1.12	0.88	1.75
Mean	317	2,446	1,906	4,060
Median	0	1,975	1,391	3,074
P20	0	957	604	1,481
P80	339	3,499	2,763	5,716
Gini	0.53	0.56	0.53	0.55

Note: These are time-averaged statistics from 2004Q1 to 2015Q2. The first two rows are in percentages, P20 and P80 imply 20th and 80th percentiles. Mean, median, P20, and P80 are real monthly income deflated by the GDP deflator, which are in units of Argentine peso.

Table 1 presents summary statistics on the income distribution across the four occupational states. The first three rows highlight significant disparities in population and income shares. While workers constitute the majority of both the population (69%) and total income (77%), the unemployed represent a sizable 10% of the population but capture a negligible 1% of income, with a median income of zero.

The primary focus of this paper is the heterogeneity within entrepreneurs (third and fourth columns), which makes up 21% of the population. The data reveals two distinct types of entrepreneurs, a distinction made clear by comparing their income shares relative to their population shares.

First, there are the self-employed, who are numerous but have low average earnings. This group makes up the vast majority of entrepreneurs (17% of the total population), yet their share of total income (15%) is smaller than their population share. Their average

income (1,906 pesos) is not only significantly lower than that of employers but also falls below the average for workers (2,446 pesos). This suggests that many individuals in this group enter self-employment out of necessity rather than for high-growth opportunities, a finding consistent with the literature on subsistence entrepreneurship.

In contrast, there are employer entrepreneurs. This group is small, making up only 4% of the population, but command a disproportionately large 7% share of total income. Their mean income (4,060 pesos) is more than double that of the self-employed. The Gini coefficients are similar across all groups, suggesting that income inequality within each occupation is comparable. However, the 20th and 80th percentiles (P20 and P80) provide insights into the lower and upper ends of the income distribution and also reveal a large income disparity between the entrepreneurial types: the 20th percentile of income for an employer (1,481 pesos) is higher than the median income for a self-employed (1,391 pesos).

This sharp divide between numerous, low-income self-employed individuals and a small cohort of high-income employers is the central empirical fact motivating the baseline model's distinction between the two groups.

Table 2: Composition of Occupations across Income Groups

Income Group	Unemployed	Workers	Self-employed	Employers
Below median	0.96	0.43	0.61	0.25
Above median	0.04	0.57	0.39	0.75

Note: Each column sums to one. Entries report the share of individuals in each occupational group whose total individual income is below or above the sample median.

To highlight the compositional heterogeneity at the heart of this paper, I examine entrepreneurial sorting by income level. Table 2 shows the distribution of each occupational group below and above median income. More than 60% of self-employed entrepreneurs are located below the median, consistent with the view that self-employment is often associated with lower-income, necessity-oriented activity. By contrast, employer entrepreneurs are concentrated in the upper half of the income distribution: 75% of employers are above the median. This pattern suggests that employer entrepreneurship is more closely associated with higher-income, opportunity-oriented activity. The same

compositional pattern emerges when using finer income disaggregations or alternative income thresholds. Capturing this compositional shift is one of the primary goals of this paper.

While the previous table focuses on the time-averaged shares of entrepreneurs across the income distribution, Table 3 presents how the population shares of occupations behave with respect to GDP across time. This presents an intriguing insight into the relationship between GDP and the population shares of entrepreneurs and workers in both the United States and Argentina. In both economies, the population share of workers are highly procyclical with respect to GDP. In the US, the correlation is 0.76 and in Argentina, the correlation is 0.86. Also, the population share of unemployed is countercyclical with respect to GDP in Argentina. There exists a large contrast between the US and Argentina when it comes to population dynamics of entrepreneurs: the population share of entrepreneurs is slightly procyclical in the US (0.24), but countercyclical in Argentina (-0.32). When the entrepreneurial group is decomposed into self-employed and employ-

Table 3: **Population Share and GDP**

	$\rho(Y, U)$	$\rho(Y, W)$	$\rho(Y, E)$	$\rho(Y, Se)$	$\rho(Y, Em)$
United States	-	0.76	0.24	-	-
Argentina	-0.85	0.86	-0.32	-0.66	0.70

Note: $\rho(Y, X)$ denotes correlation of variable X with respect to Y , where Y is GDP, U is the population share of unemployed, E is the population share of entrepreneurs (self-employed and employers combined), Se is the population share of self-employed, Em is the population share of employers. Moments using the U.S. data is taken from Kwark and Ma (2020).

ers, it can be seen that the main driver of negative correlation in Argentina is driven by self-employed entrepreneurs. This indicates that, during economic downturns, individuals turn to self-employed entrepreneurship out of necessity. However, the population share of employer entrepreneurs is highly procyclical. During economic booms, individuals may find it more optimal or less difficult to start their own businesses. The empirical findings highlight the importance of modeling occupational choice with entrepreneur heterogeneity, aggregate dynamics, and its role in shaping business cycles.

3 Baseline Model

This section develops a dynamic general equilibrium model of occupational choice. The framework is designed to be quantitatively consistent with the two key empirical facts documented above: the composition of entrepreneurs across the income distribution and the cyclicalities of their population shares.

To achieve this, the model incorporates labor market frictions to create a “push” factor into entrepreneurship out of necessity following Herreño and Ocampo (2023). The model explicitly distinguishes between self-employed and employer entrepreneurs using a similar approach as Allub and Erosa (2019). As noted in the original papers, these features are crucial for matching income and wealth inequalities within and across occupations. Furthermore, the model introduces aggregate productivity and financial shocks that drive the economy’s dynamics, which allows one to study how population dynamics influence aggregate fluctuations.

Time is discrete, and the horizon is infinite. There is a continuum of infinitely-lived households normalized to a unit measure. At the beginning of each period, households with an asset level (a) observe their skills as a worker (z_w) and as a manager (z_m). Each household is subject to uninsurable idiosyncratic shocks to their skills. Each period, households operate in one of four occupational states: unemployed (u), worker (w), self-employed entrepreneur (se), or employer entrepreneur (em). Households cannot always freely choose their preferred occupation due to labor market frictions that does not always guarantee employment (becoming a worker).

The choices available to a household at the beginning of each period depend on the realization of labor market shocks. These frictions are represented by occupation-specific probabilities of receiving job offers or being separated from a job. An unemployed agent receives a job offer with probability, p^u . If an offer is received, they become a worker. If no offer arrives, they must choose between remaining unemployed or starting a business as a self-employed or employer entrepreneur. For a worker, they face an exogenous probability of separation, p^w , which forces them into unemployment. They may also choose to quit and become an entrepreneur (self-employed or employer) any time. A self-employed

or an employer entrepreneur can choose to continue their business, or they can transition to being a worker if they receive a job offer with some exogenous probabilities p^{se} and p^{em} . They may also choose to abandon their business and become unemployed.

To capture realistic business cycle dynamics, the probabilities governing labor market transitions are modeled as a function of aggregate output of the previous period, Y_{-1} . During economic expansions when output is high, the job-finding rate increases while the job-separation rate falls, ensuring the model generates a tighter labor market in booms and a looser one in recessions, consistent with the data. Finally, employer entrepreneur hires external labor but for self-employed entrepreneurs, labor inputs must come from their own. Employers pay a fixed cost denoted by κ . Given their occupation choice, households choose their amount of final goods consumption (c) and decide how much to save with a real return denoted by r .

To summarize the timing of events, a household begins period t with assets (a_t) carried over from the previous period. At the start of the period, the household observes its current skills and the outcome of labor market shocks, which could be either job offers or separations that depends on their occupation in the prior period. Based on this information and after observing aggregate state of the economy, the household makes its occupational choice for the current period. Following this choice, production and income realization occur based on the chosen occupation: unemployed agents receive a small, fixed amount⁴, workers earn wage income, and entrepreneurs earn profits. The household also receives the gross return on its past savings, $(1 + r_t)a_t$. With all income for the period realized, the household makes its consumption-saving decision, consuming c_t and choosing savings a_{t+1} for the next period.

Preferences

Households derive utility from the consumption of a final good, denoted by c_t . Individuals maximize the expected utility of the form

⁴This can be interpreted as home production or transfers.

$$U(c_t) = \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma} - 1}{1-\gamma} \right], \quad (1)$$

where β is the discount factor between 0 and 1, and γ denotes the parameter that relates to the household's risk aversion.

Evolution of Idiosyncratic Skills

At time t , a household's skills as a worker and an entrepreneur are determined by their labor productivity z_{wt} and managerial/entrepreneurial⁵ abilities z_{mt} . Each skill is composed of a fixed component, μ_j , which determines the long-run mean of the skill, and a persistent stochastic component, x_{jt} . The skill level is given by:

$$z_{jt} = \exp(\mu_j + x_{jt}), \quad j = w, m. \quad (2)$$

The stochastic component, x_{jt} , follows a mean-zero AR(1) process.

$$x_{jt+1} = \rho_j x_{jt} + \epsilon_{jt+1}, \quad j = w, m, \quad (3)$$

where ρ_j is the persistence parameter and ϵ_{jt+1} is the innovation term of the productivity. The innovations, ϵ_{wt+1} and ϵ_{mt+1} , are drawn from a bivariate Normal distribution with a mean of zero, standard deviations σ_w and σ_m , and the correlation coefficient, ρ_{wm} .

Technology

Production in the economy takes place in two sectors: a corporate sector and a heterogeneous entrepreneurial sector. The corporate sector is standard, consisting of a representative firm that behaves competitively, taking the market wage and rental rate as given to choose its optimal capital and labor inputs. The entrepreneurial sector is populated by a continuum of agents who also take factor prices as given. A key feature of this model is the distinction between two types of entrepreneurs, which is determined by their labor

⁵I use managerial skills and entrepreneurial abilities interchangeably throughout the rest of the text.

input choices. Self-employed entrepreneurs split their time between working and managing, while employer entrepreneurs hire external labor and focus their time on managerial tasks.

Corporate Production The corporate sector produces final goods that can either be used for consumption or investment using a constant-returns-to-scale Cobb-Douglas production technology:

$$Y_t^c(K_t^c, L_t^c, Z_t) = Z_t K_t^{c,\alpha} L_t^{c,1-\alpha}, \quad (4)$$

where K_t^c and L_t^c denote capital and labor used in the corporate production. α is the share of capital used and is bounded between 0 and 1. Z_t is the exogenous aggregate productivity.

Entrepreneurial Production Following Lucas (1978) and Allub and Erosa (2019), entrepreneurs' production uses three inputs: managerial ability, labor, and capital. There are no markets for managerial ability, so entrepreneurs must only use their own managerial skills as input. The total effective managerial input is determined by the level of managerial skill z_m and time devoted to managing their businesses t_m . The production function of an entrepreneur can be written as follows:

$$Y^o(m_t, k_t, n_t; Z_t) = Z_t m_t^\nu k_t^\eta n_t^\theta, \quad o \in \{se, em\} \quad (5)$$

where $\nu + \eta + \theta = 1$ and $m_t = t_{mt} z_{mt}$ represents effective managerial supply. Likewise, k_t represents capital inputs for an entrepreneur, and n_t denotes effective units of labor input. The time allocation decision is used to differentiate two types of entrepreneurship. An entrepreneur's own labor input is written as $(1 - t_m) z_w$. The total effective labor input, n_t , is the sum of labor supplied by the entrepreneur and any external labor hired, l_t :

$$n_t = (1 - t_{mt}) z_{wt} + l_t. \quad (6)$$

Based on these choices, I define the two entrepreneurial types as follows: self-employed entrepreneurs do not hire external labor, which implies $l_t = 0$. They choose their optimal time allocation, $t_{mt} \in [0, 1]$, to split between managing and working in their own firm. On the other hand, employer entrepreneurs hire external labor ($l_t > 0$). I make the simplify-

ing assumption that they are “pure managers”, meaning they dedicate their entire time endowment to management $t_{mt} = 1$ and provide no direct labor input themselves.⁶

Finally, both types of entrepreneurs face a collateral constraint of the following form.

$$k_t \leq \phi_t a_t, \quad (7)$$

where $\phi_t \geq 1$. The functional form of this constraint is standard in the literature on financial frictions and entrepreneurship, such as Buera et al. (2015). The parameter ϕ_t captures the degree of financial development in the economy, with $\phi_t = 1$ representing pure self-financing and $\phi_t = \infty$ representing a perfect credit market. The steady state value of this parameter, $\bar{\phi}$, describes the long-run level of financial development, consistent with the literature studying misallocation over long horizons. One of the key departures in this paper is that I allow the degree of financial friction to vary over the business cycle. I model ϕ_t as a stochastic, time-varying process. This captures the idea of financial shocks, which affect firms’ ability to borrow. A large body of literature has shown this shock to be a significant driver of business cycles, such as Jermann and Quadrini (2012) and Kiyotaki and Moore (1997).

Given the market wage and interest rate (W_t, r_t) , the profit maximization problem for a self-employed entrepreneur with idiosyncratic state variables (a_t, z_{wt}, z_{mt}) can be written as

$$\pi^{se}(a_t, z_{wt}, z_{mt}; Z_t) = \max_{t_{mt}, k_t} Z_t (z_{mt} t_{mt})^\nu k_t^\eta ((1 - t_{mt}) z_{wt})^\theta - (r_t + \delta) k_t, \quad (8)$$

subject to (7). For a self-employed entrepreneur, their only choice is how much time they spend on managerial activities and the amount of capital input. For an employer entrepreneur, their profit maximization problem can be written as

$$\pi^{em}(a_t, z_{wt}, z_{mt}; Z_t, \kappa_t) = \max_{k_t, l_t} Z_t z_{mt}^\nu k_t^\eta l_t^\theta - (r_t + \delta) k_t - W_t l_t - \kappa, \quad (9)$$

subject to (7). Employer entrepreneurs must not only choose capital and time spent on managing, but they also decide how many workers to hire. Finally, κ denotes the fixed cost of production for an employer entrepreneur.

⁶The main quantitative results are robust to allowing employers to choose t_{mt} endogenously.

Aggregate Shocks Aggregate fluctuations in this economy are driven by an aggregate productivity shock, Z_t , and a financial shock, ϕ_t , which affects the tightness of the collateral constraint. The log of aggregate productivity, $\ln Z_t$, follows AR(1) process:

$$\ln Z_{t+1} = \rho_z \ln Z_t + \varepsilon_{z,t+1}. \quad (10)$$

Here, ρ_z is the persistence, and ε_{zt} is the innovation term that follows a normal distribution with a standard deviation σ_z . Similarly, financial shock also follows an AR(1) process in logs.

$$\ln \phi_{t+1} = (1 - \rho_\phi) \ln \bar{\phi} + \rho_\phi \ln \phi_t + \varepsilon_{\phi,t+1}, \quad (11)$$

where $\bar{\phi}$ is the long-run level of financial development, ρ_ϕ is the persistence of financial shocks, and the innovation term, $\varepsilon_{\phi,t}$, follows a normal distribution with a standard deviation σ_ϕ . I further assume that the two shocks are correlated, which is represented by $\rho_{z,\phi}$.

Household Problem

The household's problem can be formulated as a dynamic programming problem. At the beginning of a period, the household enters with assets and observes its current skills. Let $s = (a, z_w, z_m)$ denote the household's individual state, where a is assets, z_w is worker productivity, and z_m is managerial ability. In addition, the household enters the period with a lagged occupation $o \in \{u, w, se, em\}$ which is the occupation chosen in the previous period. This lagged occupation determines the type of labor market shocks faced by households at the beginning of period t . After the labor market shock is realized, the household chooses its current occupation j , production takes place, and realizes income. The aggregate state of the economy is $S = (\Lambda^-, Z, \phi, Y_{-1})$, where Λ^- is the beginning-of-period distribution of households, Z is aggregate productivity, ϕ is the financial condition of the economy, and Y_{-1} is the aggregate output in the previous period that determines the state of the labor market today.

Given the timing, it is useful to define two value functions. Let $\tilde{V}^o(s, S)$ be the beginning-

of-period value for a household whose occupation in the previous period was o . Furthermore, let $v^j(s, S)$ denote the value of choosing occupation j in the current period, after the aggregate state has been realized and the feasible set of occupations is known. The current occupation j determines current income and, through the budget constraint, the household's consumption-saving decision. It also becomes the household's lagged occupation in the next period.

Consider first a household that was unemployed in the previous period, ($o = u$). At the beginning of the current period, the household receives a job offer with probability $p^u(S)$.⁷ If the offer arrives, the household becomes a worker in the current period. If no offer arrives, the household cannot choose wage work and must choose among unemployment, self-employment, and employer entrepreneurship. Therefore, the beginning-of-period value for households who were unemployed last period can be written as:

$$\tilde{V}^u(s, S) = \left[p^u(S)v^w(s, S) + (1 - p^u(S)) \max_{j \in \{u, se, em\}} \{v^j(s, S) + \varepsilon(j)\} \right]. \quad (12)$$

The first term is the value of receiving a job offer and working in the current period. The second term is the value of not receiving an offer, in which case the household chooses from the restricted set of feasible occupations. The term $\varepsilon(j)$ is an occupation-specific preference shock drawn from a Type-I Extreme Value distribution.⁸ The inclusion of this shock means that occupational choice is probabilistic. This allows the value function to be differentiable with respect to the aggregate state variables.

For a household that was a worker in the previous period, ($o = w$), it faces an exogenous separation shock with probability $p^w(S)$. If the household is separated, the agent is forced into unemployment in the current period and receives the value $v^u(s, S)$. If the household is not separated, the agent can continue as a worker or switch to any other occupation. The value function can be written as:

$$\tilde{V}^w(s, S) = \left[p^w(S)v^u(s, S) + (1 - p^w(S)) \max_{j \in \{u, w, se, em\}} \{V^j(s, S) + \varepsilon(j)\} \right], \quad (13)$$

⁷In the quantitative implementation, labor-market probabilities are functions of lagged aggregate output, which is known at the beginning of the period before current production takes place. For notational simplicity, I write these probabilities as functions of the aggregate state S .

⁸The smoothing parameter that controls the discrete-choice probabilities is set to 0.015, which is a standard value in the literature.

where the second term is the expected value of choosing from the full set of occupations.

For a household that was self-employed or an employer entrepreneur in the previous period, $o \in \{se, em\}$, the household receives a job offer with probability $p^o(S)$. If an offer is received, the agent can either choose to be a worker or continue as the same type of entrepreneur. If an offer is not received, the agent can continue to operate the business as its current type, exit to unemployment, or switch entrepreneurial type. The value function is

$$\tilde{V}^o(s, S) = \left[p^o(S) \max_{j_1 \in \{w, o\}} \{v^{j_1}(s, S) + \varepsilon(j_1)\} + (1 - p^o(S)) \max_{j_2 \in \{u, se, em\}} \{v^{j_2}(s, S) + \varepsilon(j_2)\} \right], \quad (14)$$

where $o \in \{se, em\}$ and j_1 and j_2 represent the sets of occupational choices available to entrepreneurs with and without job offers, respectively.

After facing their labor market shocks at the beginning of the period and observing the aggregate state of the economy, households choose their current occupation j . For a given occupation j , the value function is written as:

$$v^j(s, S) = \max_{c, a'} \left\{ \frac{c^{1-\gamma} - 1}{1 - \gamma} + \beta \mathbb{E}[\tilde{V}^j(s', S') | s, S] \right\}, \quad (15)$$

subject to the budget constraint:

$$c + a' = inc^j(s, S) + (1 + r(S))a, \quad (16)$$

$$a' \geq 0. \quad (17)$$

Here, $inc^j(s, S)$ is the income generated from the current occupation choice j . Unemployed agents receive a small fixed amount $\underline{b}$. Workers earn wage income which depends on the market wage rate and their labor productivity, $W(S)z_w$. Entrepreneurs earn profits $\pi^j(s, S)$, which depend on their occupation and skill. In addition, the household receives the gross return $(1 + r(S))a$ on assets carried into the period. Finally, the expectation $\mathbb{E}[\cdot]$ is taken over the realization of next period's skill and aggregate state S , which depend on their current occupation.

3.1 Market Clearing and Recursive Competitive Equilibrium

Aggregate consumption (C_t) and output (Y_t) are the sum of their individual and sectoral components.

$$C_t = \int c_{it} d\Lambda_t, \quad (18)$$

$$Y_t = Y_t^c + Y_t^{se} + Y_t^{em}, \quad (19)$$

where Y_t^c is the output of the corporate sector, and Y_t^{se} and Y_t^{em} are the aggregate outputs of self-employed and employers, respectively. $d\Lambda_t$ is the distribution of households after occupational choices are realized. Y_t^{se} and Y_t^{em} are obtained by integrating the output of individual entrepreneurs over their respective distributions, $d\Lambda_t^{se}$ and $d\Lambda_t^{em}$. This can be written as

$$Y_t^{se} = \int y_{it}^{se} d\Lambda_t^{se}, \quad Y_t^{em} = \int y_{it}^{em} d\Lambda_t^{em}. \quad (20)$$

Aggregate capital is defined as

$$K_t = K_t^c + \int k_{it}^{se} d\Lambda_t^{se} + \int k_{it}^{em} d\Lambda_t^{em} = K_t^c + K_t^{se} + K_t^{em}, \quad (21)$$

I define L_t as the total labor demanded by the corporate sector and employer entrepreneurs:

$$L_t = L_t^c + \int l_{it} d\Lambda_t^{em} = L_t^c + L_t^{em}. \quad (22)$$

Self-employed labor is used internally within self-employed firms and is not traded in the market. Finally, in the model with household heterogeneity and financial frictions, the aggregate productivity of entrepreneurial sectors is made up of exogenous component, Z_t , and an endogenous component, A_t . The endogenous component captures the sector's allocative efficiency, which is determined by the degree of misallocation and the joint distribution of skills, assets, and choices of households. I define and measure the endogenous productivity as follows:

$$A_t^{se} = Y_t^{se} / (Z_t (K_t^{se})^\eta), \quad (23)$$

$$A_t^{em} = Y_t^{em} / (Z_t (K_t^{em})^\eta (L_t^{em})^\theta). \quad (24)$$

In equilibrium, the labor demanded by the corporate sector and employer entrepreneurs must equal the labor supplied by workers. Similarly, the amount of capital demanded by the corporate sector and entrepreneurs must equal the savings done by households.

$$\int z_{wt} d\Lambda_t^w = L_t, \quad (25)$$

$$\int a_{it} d\Lambda_t = K_t, \quad (26)$$

The above market clearing conditions lead to the final goods market clearing condition, which can be stated as follows:

$$C_t + I_t = Y_t - \kappa \int d\Lambda_t^{em}, \quad (27)$$

where the aggregate investment I_t is defined as $K_{t+1} - (1 - \delta)K_t$. The last term is the deadweight loss that stems from the fixed cost of production for households that choose to be employer entrepreneurs.⁹

A *recursive competitive equilibrium* for this economy is given by a set of pricing functions $\{W(S), r(S)\}$, corporate capital and labor decisions $\{K^c(S), L^c(S)\}$, households' policy functions $\{c(s, S), a'(s, S), j(s, S), t_m(s, S), l(s, S), k(s, S)\}$, value functions $v(s, S)$, and law of motion for the distribution $\Gamma(\cdot)$ such that the following conditions hold:

1. Households maximize their value functions by choosing policy functions and their choices of occupations given prices;
2. Corporate firms maximize profit by choosing capital and labor given prices;
3. Labor market, capital market, and goods market clear;
4. Aggregate law of motion is generated by savings decisions by households.

4 Solution Method and Calibration

This section outlines the computational strategy used to solve the model and the calibration procedure used to discipline its parameters.

⁹While there is also home production income from unemployed households, they are negligible in the aggregate and is thus ignored.

4.1 Solution Method

The model features both rich household heterogeneity and aggregate shocks, making it computationally complex. A standard method for solving such models is the algorithm proposed by Krusell and Smith (1998), which involves simulating the economy to approximate the law of motion for the distribution of wealth. However, this simulation-based approach is computationally intensive and can become infeasible when multiple shocks or a large state space are present.

Given that the paper’s focus is on business cycle properties, I follow the perturbation approach recommended by Reiter (2009). This method involves first solving the non-linear model in the stationary equilibrium (without aggregate shocks) and then linearizing the equilibrium dynamics with respect to the aggregate shocks. As shown by Bayer and Luetticke (2020) and others, this approach provides a good balance of accuracy and computational speed for analyzing business cycle questions. Specifically, I solve the household’s problem using value function iteration on a discretized state space and then linearize the full system to compute second moments. Appendix C provides details on how the model is solved and simulated.

4.2 Calibration

The calibration is done in two separate steps. The goal is to match occupation-level statistics from the data, as well as key aggregate business cycle moments. First, I externally calibrate a subset of parameters based on standard values in the literature and estimates from the worker income process. Second, I internally calibrate the remaining parameters in two stages. In the first stage, I target moments in the stationary equilibrium, calibrating parameters that primarily affect outcomes in the steady state. In the second stage, using the linearized model, I calibrate the parameters governing the aggregate shock processes to match key business cycle moments. The data for this procedure come from two main sources. Micro-level moments on occupational choice and income distribution are calculated using Argentina’s Permanent Household Survey (EPH) from 2004Q1 to 2015Q2. Aggregate business cycle moments are from the International Financial Statistics (IFS)

Table 4: **Externally Calibrated Parameters**

Parameter	Description	Value
γ	Risk aversion	2
β	Discount factor	0.98
α	Capital share in corporate production	0.32
ν	Managerial share in entrepreneur production	0.20
η	Capital share in entrepreneur production	0.32
θ	Labor share in entrepreneur production	0.48
δ	Capital depreciation rate	0.025
$\underline{b}$	Unemployed income	0.001
ρ_w	Persistence of working abilities	0.96
σ_w	S.d. of worker ability innovations	0.13

database.

External Calibration

Table 4 presents the externally calibrated parameters. The risk aversion parameter, γ , is set to 2, a standard value in the literature. I set the annualized world interest rate to 2% and choose the discount factor, β , so that the stationary equilibrium is consistent with this interest rate, which leads to a value of about 0.98. Production parameters are also chosen to be consistent with the previous literature. I set the capital share in corporate production, α , to 0.32. The managerial, capital, and labor shares in entrepreneurial production, (ν, η, θ) , are set to 0.20, 0.32, and 0.48, respectively, based on the parameters in Guner et al. (2008). The quarterly capital depreciation rate is set to 0.025, which is a standard value in the literature. Finally, I estimate the persistence and innovation standard deviation of worker ability to match the income process of workers in the microdata, which are $\rho_w = 0.96$ and $\sigma_w = 0.13$, respectively. Appendix A.3 provides details on the estimation procedure.

Internal Calibration

I internally calibrate the remaining parameters in two stages. First, I calibrate parameters that primarily affect the stationary equilibrium. Second, using the linearized model, I calibrate the parameters governing the aggregate shock processes to match key business cycle moments.

For the stationary equilibrium, there are 11 parameters calibrated to match 12 empirical moments. While all parameters affect all moments jointly, it is helpful to identify which parameters have the greatest influence on specific targets. Five parameters related to the idiosyncratic productivity processes remain to be calibrated. The productivity processes, z_w and z_m , are discretized into a finite-state Markov chain using the Tauchen method. The persistence parameter of managerial ability, ρ_m , and the correlation coefficient, $\rho_{w,m}$, are disciplined primarily by the relative average incomes across occupations: worker/employer, self-employed/worker, and self-employed/employer. The standard deviation of innovations to managerial ability, σ_m , is targeted to match the variance of log income for entrepreneurs. The fixed cost of production, κ , and the long-run means of the productivity processes, μ_w and μ_m , are used to target the population shares of workers, self-employed entrepreneurs, and employer entrepreneurs. Furthermore, I set the steady state collateral constraint parameter, $\bar{\phi}$, to match an external credit-to-GDP ratio of 25%.

Finally, I assume that the labor market processes, p^j follow a logistic distribution. This is expressed as

$$p^j(Y_{t-1}) = \frac{\exp(\psi^j(\ln Y_{t-1} - \ln \bar{Y}) + \tilde{p}^j)}{1 + \exp(\psi^j(\ln Y_{t-1} - \ln \bar{Y}) + \tilde{p}^j)}, \quad (28)$$

where $j = u, w, se, em$ and $\tilde{p}^j$ governs the steady-state level of the probability for each occupation j . The four steady-state level parameters, $\tilde{p}^j$, are then internally calibrated to match key moments of occupational flows in the data. While I estimate $\tilde{p}$ in practice, I report p^j for easier interpretations.

I capture the cyclical nature of the labor market in a parsimonious way. The labor market probabilities depend on the log-deviations of lagged aggregate output, Y_{t-1} , from its steady-state value, $\bar{Y}$. The sensitivity of these probabilities to the business cycle is governed by a single parameter, $\psi > 0$. I assume that all job-finding/offer probabilities move together with the business cycle, and the job-separation probability moves countercyclically. This implies that $\psi^u = \psi^{se} = \psi^{em} = \psi$ and $\psi^w = -\psi$.

After linearizing the model with respect to aggregate shocks, there are 6 parameters chosen to match 6 moments from aggregate dynamics. The persistence and standard deviation of TFP shock, ρ_z and σ_z , are chosen to match the serial correlation and standard

Table 5: **Internally Calibrated Parameters**

Parameter	Description	Value
Stationary Equilibrium		
ρ_m	Persistence of managerial abilities	0.92
σ_m	S.d. of managerial ability innovations	0.60
$\rho_{w,m}$	Correlation between working and managing	0.75
μ_w	Shift parameter of working abilities	0.30
μ_m	Shift parameter of managerial abilities	0.13
κ	Fixed cost of production	0.07
$\bar{\phi}$	Collateral constraint parameter	1.24
p^u	Job-finding rate of unemployed	0.43
p^w	Job-separation rate of workers	0.09
p^{se}	Job-finding rate of self-employed	0.22
p^{em}	Job-finding rate of employers	0.36
Aggregate Parameters		
ρ_Z	Persistence of TFP shock	0.71
σ_Z	S.d. of TFP shock	0.03
ρ_ϕ	Persistence of financial shock	0.89
σ_ϕ	S.d. of financial shock	0.08
$\rho_{z,\phi}$	Correlation between TFP and financial shock	0.67
ψ	Labor market parameter	1.54

Note: This table lists parameter values that are chosen to match the moments in Table 6.

deviation of aggregate output. The persistence and standard deviation of financial shock, ρ_ϕ and σ_ϕ , are chosen to match the serial correlation and standard deviation of income of entrepreneurs (including both employers and self-employed). The correlation of the shock $\rho_{z,\phi}$ to match the cyclicity of employer shares, and ψ is used to match the standard deviation of the population share of workers.

Table 5 presents the resulting parameter values from the internal calibration. The persistence of managerial ability is high, with $\rho_m = 0.92$. The innovation standard deviation of managerial ability is much larger than that of worker ability, with $\sigma_m = 0.60$ compared to $\sigma_w = 0.13$. This higher dispersion helps the model generate the higher income variance observed among entrepreneurs. The innovations to the two skill processes are also highly and positively correlated, with $\rho_{w,m} = 0.75$. Although the shift parameter for worker ability is higher than that for managerial ability, the larger dispersion of managerial ability generates a thick upper tail, which is important for matching the income of employer

entrepreneurs.

The model also includes several parameters related to various frictions. The fixed cost for employer entrepreneurs, κ , is calibrated to 0.07, which represents approximately 16% of total employer output. In terms of the labor market frictions, the job-finding rates are highest for unemployed individuals with the value of 43 percent, followed by employers with the value of 0.36, and lower for the self-employed ($p^{se} = 0.22$).

Finally, the parameters for the aggregate TFP shock process are calibrated to match the business cycle properties of the Argentinian GDP. The persistence of the shock, ρ_Z , is estimated to be 0.71 and its standard deviation, σ_Z , is 0.03. These values are consistent with estimates for aggregate productivity shocks in the representative-agent business cycle literature. After calibration, I find that the financial shock is much more volatile and persistent than the TFP shock. The innovations to the two shocks are found to be strongly and positively correlated. The parameter governing the cyclicalities of the labor market, ψ , is estimated to be 1.54. This implies that a 1% increase in aggregate output above its steady state leads to a 0.4 percentage point increase in the job-finding probability for an unemployed individual.

Table 6 presents the targeted moments and compares the model's performance to the data. The results are organized into two panels: moments related to the stationary equilibrium and moments related to aggregate dynamics. Overall, the model does a very good job of capturing the key features of the data. In the stationary equilibrium, the model matches the population shares across occupations, the average income ratios across occupations, and the income dispersion among entrepreneurs.

The model also replicates the occupational composition of the economy. The calibrated population shares of unemployed individuals, workers, self-employed entrepreneurs, and employer entrepreneurs closely match their empirical counterparts. Furthermore, the model matches the average income ranking across these groups, with employers earning more than workers and workers earning more than the self-employed. The calibrated income ratios also align well with the data. The model also successfully matches the targeted moments related to aggregate dynamics.

To externally validate the model, I compare key untargeted moments from the simu-

Table 6: **Calibration Results**

Moment	Data	Model
Stationary Equilibrium		
Population shares		
Unemployed	0.10	0.09
Self-employed	0.17	0.18
Workers	0.69	0.70
Employers	0.04	0.03
Average income of X to the average income of Y		
Self-employed/workers	0.78	0.77
Workers/employers	0.60	0.60
Self-employed/employers	0.47	0.46
Variance of log income of entrepreneurs	0.94	0.94
Transition into employment from X		
Unemployed	0.42	0.43
Workers	0.89	0.90
Self-employed	0.20	0.18
Employers	0.10	0.11
External credit-to-GDP ratio	0.25	0.25
Aggregate Dynamics		
Serial correlation of output	0.83	0.83
Standard deviation of output	0.05	0.05
Serial correlation of entrepreneur income	0.69	0.69
Standard deviation of entrepreneur income	0.07	0.07
Correlation of employer share with GDP	0.70	0.70
Standard deviations of worker share	0.01	0.01

Table 7: **Untargeted Moments**

Moment	Data	Model
Income Shares		
Unemployed	0.01	0.01
Workers	0.77	0.79
Self-employed	0.15	0.15
Employers	0.07	0.05
Composition of Entrepreneurs		
Below median		
Share of self-employed	0.61	0.64
Share of employers	0.25	0.20
Above median		
Share of self-employed	0.39	0.36
Share of employers	0.75	0.80

lation with the data, as shown in Table 7. I focus on two sets of statistics. First, I compare the income shares held by each occupation. These moments are closely matched by the model. Second, I show that the model successfully replicates the sorting of entrepreneurial types across the income distribution, even though these moments were not directly targeted.

In the data, 61% of self-employed entrepreneurs are below the median income, whereas only 25% of employers are below the median. The model generates corresponding shares of 64% and 20%, respectively. Conversely, 39% of self-employed entrepreneurs are above the median, whereas 75% of employers are above the median. In the model, these shares are 36% for self-employed entrepreneurs and 80% for employers, which are close to the data.

This result indicates that the model's core sorting mechanisms are working as intended. The fixed cost, κ , acts as a significant barrier to entry into employer entrepreneurship, while the self-employed face limitations in scaling because they cannot hire external labor. Only individuals with a combination of high managerial ability and sufficient wealth, both of which are correlated with high income, can profitably overcome this barrier. This mechanism endogenously generates an economy in which necessity-driven self-employment is more prevalent below the median of the income distribution, while opportunity-driven employer entrepreneurship is more prevalent above the median.

It is important to note that the model, consistent with the data, still predicts that some employers are located below the median of the income distribution. This occurs because idiosyncratic shocks to managerial ability are persistent. For a small subset of individuals, such as those with sufficient wealth and managerial productivity but low worker ability, it remains optimal to choose employer entrepreneurship because they are more likely to overcome the fixed cost. However, their income after paying the fixed cost can remain relatively low. Conversely, some self-employed entrepreneurs optimally continue in self-employment because their managerial ability is not sufficiently high relative to their worker ability.

Next, I evaluate the model's performance on the quarterly occupational transition rates shown in Table 8. The model was calibrated only to match the moments related

Table 8: Occupational Transition Rates

	Data	Model		Data	Model		Data	Model		Data	Model
$U \rightarrow U$	0.37	0.20	$U \rightarrow W^*$	0.42	0.43	$U \rightarrow Se$	0.19	0.35	$U \rightarrow Em$	0.01	0.02
$W \rightarrow U$	0.04	0.09	$W \rightarrow W^*$	0.89	0.90	$W \rightarrow Se$	0.06	0.01	$W \rightarrow Em$	0.01	0.00
$Se \rightarrow U$	0.08	0.04	$Se \rightarrow W^*$	0.20	0.18	$Se \rightarrow Se$	0.64	0.75	$Se \rightarrow Em$	0.08	0.03
$Em \rightarrow U$	0.02	0.02	$Em \rightarrow W^*$	0.10	0.11	$Em \rightarrow Se$	0.26	0.20	$Em \rightarrow Em$	0.62	0.67

Note: Values represent quarter-to-quarter transition probabilities between occupational states. An asterisk (*) indicates a moment that was explicitly targeted during the model’s calibration.

to transitions into, or persistence in, wage employment, which are denoted by asterisks. Despite not targeting the remaining transition rates directly, the model captures the general patterns of occupational mobility reasonably well.

However, there are some notable deviations. Quantitatively, the model understates the persistence of unemployment, with 20% of unemployed individuals remaining unemployed in the model compared to 37% in the data. It also overstates entry into self-employment from unemployment, with a transition rate of 35% in the model compared to 19% in the data. Furthermore, the model generates a higher job-separation rate from wage employment than in the data, with $W \rightarrow U$ equal to 9% in the model compared to 4% in the data. This discrepancy arises because the calibration of the separation rate is disciplined by the persistence of the worker state, $W \rightarrow W$, rather than by the direct transition from wage employment into unemployment.

Nonetheless, the model replicates the fact that wage employment and both types of entrepreneurship are persistent in the data. In particular, the model generates high persistence among workers and employer entrepreneurs, and it captures broad occupational mobility patterns despite not directly targeting most of the transition rates.

5 Aggregate Dynamics with Occupational Heterogeneity

With the model calibrated and validated on Argentinian data, this section uses it as a laboratory to address the paper’s central question: what are the business cycle implications of financial development policies? This question cannot be answered with reduced-form evidence alone. Financial development is structural: it alters entry, scale, and hiring de-

cisions and shifts equilibrium wages and returns in general equilibrium. Because these behavioral and price responses are endogenous, the mapping from credit conditions to aggregate dynamics is not policy invariant (Lucas critique). A quantitatively disciplined model is therefore needed to run the relevant counterfactual and to separate within-sector stabilization from the composition (and covariance) effects created by reallocating activity toward employer entrepreneurs.

I begin by showing that the model matches key aggregate business cycle statistics and replicates the observed dynamics of population shares and occupational incomes across groups. I then study the model’s baseline transmission mechanisms using impulse responses to the aggregate shocks. Finally, with the validated framework, I analyze the business cycle consequences of broad-based financial development by varying the steady state collateral constraint, $\bar{\phi}$. The main finding is that the relationship between financial development and aggregate volatility is positive.

5.1 Second Moments

I now evaluate the model’s ability to replicate key business cycle statistics from the data, as shown in Table 9. The table is organized into three panels: aggregate variables, population shares, and occupational total incomes.

The model successfully captures the high volatility of investment relative to output, a standard feature of business cycles. However, it does not generate the “excess consumption volatility” commonly observed in emerging markets. In the data, consumption is slightly more volatile than output (a ratio of 1.09), whereas the model produces consumption that is about half as volatile as output (a ratio of 0.54). Resolving this well-known puzzle is beyond the scope of this paper’s central research questions. The discrepancy likely stems from the model’s shock structure; as argued by Aguiar and Gopinath (2007), a more persistent TFP shock that behaves like a permanent income shock would be needed to generate a stronger consumption response.¹⁰

¹⁰Other papers, such as Hong (2023), find that higher MPCs in emerging markets contribute to excess consumption volatility. However, I do not pursue this due to data limitations on household-level consumption.

Table 9: Business Cycle Statistics: Model vs. Data

	Relative Std. Dev.		Correlation w/ Output	
	Data	Model	Data	Model
Aggregate variables				
Output	1.00*	1.00*	1.00	1.00
Consumption	1.09	0.54	0.95	0.79
Investment	2.39	2.95	0.96	0.94
Population shares				
Unemployed	0.22	0.12	-0.85	-0.91
Worker	0.24*	0.24*	0.86	0.74
Self-Employed	0.11	0.20	-0.66	-0.65
Employer	0.07	0.09	0.70*	0.70*
Occupational incomes				
Unemployed	3.14	4.30	0.43	0.59
Worker	1.61	1.19	0.73	0.99
Self-employed	0.85	1.04	0.73	0.23
Employer	3.27	4.87	0.51	0.82

Note: All variables, except population shares, are logged. Standard deviations are expressed relative to the standard deviation of output. All variables are linearly filtered in the data while the model reports theoretical moments. Data moments are calculated using quarterly Argentinian data from 2004Q1 to 2015Q2. An asterisk (*) indicates a moment that was explicitly targeted during the model's calibration.

The model performs well in matching the cyclical patterns of occupational shares and incomes, even though most of these moments were not directly targeted. In terms of population shares, the model correctly generates that the shares of unemployed and self-employed individuals are countercyclical, while the shares of workers and employers are procyclical. The model's correlations are quantitatively close to the data, particularly for the untargeted unemployed (-0.91 in the model vs. -0.85 in the data) and self-employed (-0.65 in the model vs. -0.66 in the data) groups. This success is driven by the model's realistic labor market frictions, where an economic boom (and the associated rise in wages) endogenously pulls individuals out of unemployment and self-employment and into wage work or to pursue employer entrepreneurship.

In terms of aggregate occupational incomes, the model captures the key volatility rankings. Consistent with the data, employer and unemployed incomes are the most volatile, while self-employed income is the most stable. The model also generates the

procyclicality of incomes for all productive groups. These endogenous dynamics are a key strength of the framework: even though all entrepreneurs face the same aggregate shocks, their heterogeneous production structures and financial constraints cause them to respond differently. Overall, the model provides a good qualitative and quantitative match to the second moment properties of the Argentinian data, capturing the key relative volatilities and cyclical patterns that are central to the paper's mechanism.¹¹

5.2 Impulse Response Analysis

This section analyzes the model's dynamic properties by examining the impulse responses to a one percent positive shock to aggregate productivity (Z_t) and financial shock (ϕ_t). To highlight the role of entrepreneurial heterogeneity, the analysis focuses on the differential responses of the corporate, self-employed, and employer sectors.¹² Unless otherwise noted, all variables are shown as percent deviations from their steady state values.

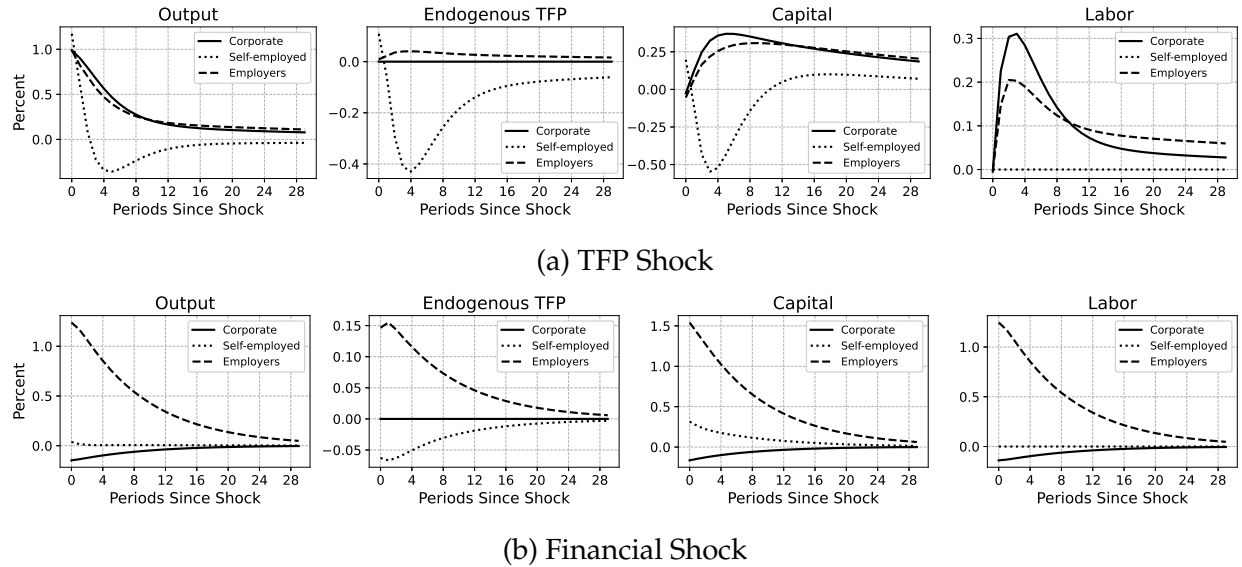


Figure 1: Impulse Response Analysis

Note: These figures show the impulse response of sectoral output, endogenous TFP, capital, and labor to a unit positive shock. All variables are in percent deviations from the steady state.

¹¹Appendix E presents a sensitivity analysis with respect to key parameters.

¹²The impulse responses for other key variables, such as aggregate quantities and population shares, are presented in the appendix.

The top panel of Figure 1 shows the impulse responses to a positive TFP shock. As the shock hits, output rises across all three production sectors, but the responses differ significantly in magnitude and persistence. The corporate and employer sectors experience a strong and persistent increase in output, rising by 1% on impact. In contrast, while the response of the self-employed sector rises by more than 1 percent on impact, it is much more transitory, even turning negative after two quarters.

The underlying factor responses reveal the mechanisms driving these heterogeneous output dynamics. The endogenous TFP of the self-employed sector actually declines, counteracting the direct effect of the aggregate TFP shock. This is driven by the nature of “necessity” entrepreneurship that characterizes this group. The productivity boom makes wage work a more attractive outside option (due to higher wages and job-finding rates), leading to an overall lower share of self-employed entrepreneurs. Those with high working abilities become employees, while those with strong managerial skills become employers. This compositional shift lowers the productivity of the remaining self-employed.

In contrast, the endogenous TFP of the employer sector rises moderately. This reflects the “opportunity-driven” nature of this group. The favorable economic conditions create profitable opportunities, allowing high-quality entrepreneurs to enter and scale up their operations. These divergent TFP responses, combined with a larger increase in capital and labor for employers, explain the differential output dynamics between the two sectors. The factor responses further highlight this mechanism: capital and labor demand from the corporate and employer sectors rise persistently, while capital demand from the self-employed sector falls after two quarters.

The bottom panel of Figure 1 shows the responses to a positive financial shock (a loosening of the collateral constraint). The transmission mechanism of this shock is fundamentally different. It has no direct effect on the corporate sector, but instead operates by relaxing the borrowing constraints of entrepreneurs.

The results show that the employer sector is the primary beneficiary and transmitter of the financial shock. Employer output rises by approximately 1.2% on impact, driven by a substantial increase in both capital and labor inputs. The self-employed sector is unable to translate the positive financial shock into a large output gain due to its smaller

scale and more limited use of capital (from lack of external labor), ultimately leading to a slight decline of endogenous productivity. Thus, the financial shock is propagated to the rest of the economy primarily through the higher capital and labor decisions of employer entrepreneurs, which causes a reallocation of capital and labor away from the corporate sector.¹³

This analysis highlights a key finding: the two entrepreneurial sectors respond to the same aggregate shocks differently. The self-employed sector's response is dampened by the higher switch rate towards better opportunities in response to TFP shocks and limited by its scale in response to financial shocks. The employer sector, in contrast, is able to leverage its larger scale and access to hired labor to amplify both types of shocks. This differential propagation mechanism is central to the paper's main result. Financial development alters the composition of the entrepreneurial sector in the steady state and shifts the economy's weight from the less responsive self-employed to the more responsive employers, which fundamentally changes the economy's aggregate volatility.

5.3 Financial Development and Business Cycles

I now use the validated model to conduct the main counterfactual analysis on the relationship between financial development and aggregate dynamics. A large body of literature has shown that in steady state models, financial development improves resource allocation, leading to higher aggregate TFP, output, and welfare (e.g., Buera et al. (2015); Buera et al. (2011)). However, the implications of such policies for business cycles are less well understood. This section contributes by analyzing these dynamic consequences, moving beyond a purely steady state focus.

It is important to emphasize that the quantitative experiments in this paper isolate the effects of a pure credit expansion. In practice, episodes of financial deepening are rarely so simple. Historical evidence shows that credit booms without improvements in prudential regulation or institutional quality are often linked to heightened volatility and financial fragility (e.g., Ranciere et al. (2006) and Reinhart and Rogoff (2009)).

¹³Appendix D presents additional impulse responses of aggregate variables, population shares, and group-level incomes to these shocks.

By contrast, the reductions in macroeconomic volatility observed in cross-country data generally arise when credit deepening occurs alongside broader regulatory reforms and institutional strengthening (e.g., Denizer et al. (2002) and Cerutti et al. (2017)). The model intends to capture a specific scenario of financial development: one in which access to credit expands, but the structure and volatility of aggregate shocks remain unchanged.

Formally, financial development is represented by an increase in the steady-state value of the collateral constraint parameter, $\bar{\phi}$. I re-solve the stationary equilibrium for values of $\bar{\phi}$ ranging from 1 to 3 and then linearize around the new steady states to compute business-cycle moments. The analysis first considers steady-state outcomes and then turns to second-moment properties.

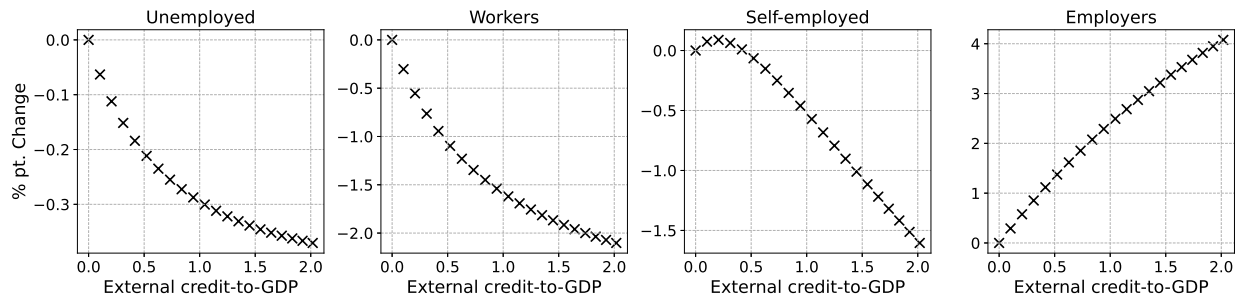


Figure 2: Long-Run Impact of Financial Development: Population Shares

Note: The figure plots the deviation of the steady-state population share for each occupation across values of the collateral constraint parameter. The x-axis shows the external credit-to-GDP ratios implied by each value of the collateral constraint parameter. The y-axis shows the change in percentage points relative to the completely self-financed economy.

Given the focus on occupational heterogeneity, I first examine how financial development alters the long-run distribution of the population. Figure 2 plots the change in the population share of each occupation as a function of the external credit-to-GDP ratio. The figure shows that financial development increases entrepreneurship overall, but its effects differ across entrepreneurial types. The response of self-employment is non-monotonic. At very low levels of financial development, the self-employed share rises slightly, by less than 0.1 percentage points. This initial increase reflects the fact that small relaxations of borrowing constraints allow some households to operate small-scale businesses without hiring outside labor or paying the fixed cost required for employer entrepreneurship. As financial development deepens, however, the employer margin becomes more attractive,

and productive self-employed entrepreneurs can scale up, hire labor, and become employers. As a result, when the external credit-to-GDP ratio reaches 2, the self-employed share is about 1.6 percentage points below its level in the self-financed economy. By contrast, the share of employer entrepreneurs rises monotonically, increasing by more than 4 percentage points, as individuals with high managerial abilities gain access to more capital and the return to operating larger firms increases. Thus, financial development does not simply increase the number of entrepreneurs; it reallocates entrepreneurial activity from small-scale self-employment toward larger-scale employer entrepreneurship.

The shares of workers and the unemployed also decline as entrepreneurship becomes a more attractive outside option. The worker share falls by more than 2 percentage points, while the unemployed share decreases by about 0.35 percentage points. This reflects that, as financial development raises the return to entrepreneurial activity, individuals at the margin with sufficient wealth and managerial ability transition out of wage work or unemployment into entrepreneurship, especially employer entrepreneurship.

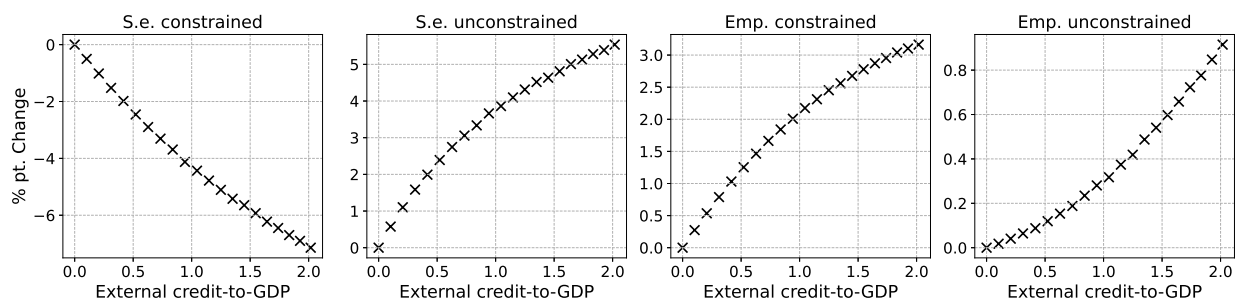


Figure 3: Long-Run Impact of Financial Development: Constrained and Unconstrained Shares

Note: The figure plots the deviation of the steady-state population share for each entrepreneurial subgroup. The y-axis shows the change in percentage points relative to the self-financed economy.

To further understand the population shifts, I decompose the entrepreneurial population into financially constrained and unconstrained subgroups. Figure 3 plots the change in population shares for each subgroup. The primary effect of financial development is a broad reallocation away from constrained self-employment toward all other entrepreneurial categories. As the collateral constraint loosens, the share of constrained self-employed individuals falls sharply by roughly 7 percentage points while the shares

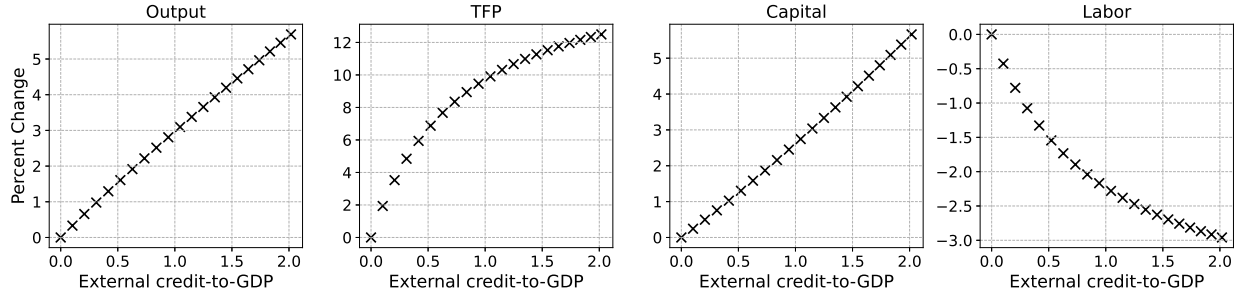


Figure 4: Long-Run Impact of Financial Development: Supply-side Aggregates

Note: The figure plots the percent deviation of steady-state aggregate output, TFP (sum of endogenous TFPs across sectors), capital, and labor as a function of external credit-to-GDP. Deviations are calculated as percent changes relative to $\bar{\phi} = 1$.

of unconstrained self-employed and both types of employers increase monotonically.

Interestingly, the share of constrained employer entrepreneurs rises because financial development enables the most productive self-employed to transition into employer entrepreneurship. However, as they are often individuals with relatively low wealth, they immediately become financially constrained in their new roles. This finding reveals that financial development not only shifts the composition across types but also alters the nature of financial constraints within them. Lastly, it must be noted that in the aggregate, the total share of constrained entrepreneurs declines by about 3.9 percentage points, implying that improved access to credit does reduce the number of financially constrained entrepreneurs.

Next, I turn to long-run supply-side aggregates: output, TFP, capital, and labor. Figure 4 plots their percentage deviations relative to the completely self-financed economy. Consistent with the literature, both output and capital rise strongly with financial development, both increasing by approximately 6 percent.

The response of aggregate labor is less direct because financial development affects both labor demand and occupational sorting. On the one hand, higher wages and increased labor demand by employer entrepreneurs tend to expand labor supply. On the other hand, higher returns to entrepreneurship, together with reduced labor demand from the corporate sector, put downward pressure on total labor supply. In equilibrium, the second force modestly dominates, so aggregate labor declines slightly. This decline is

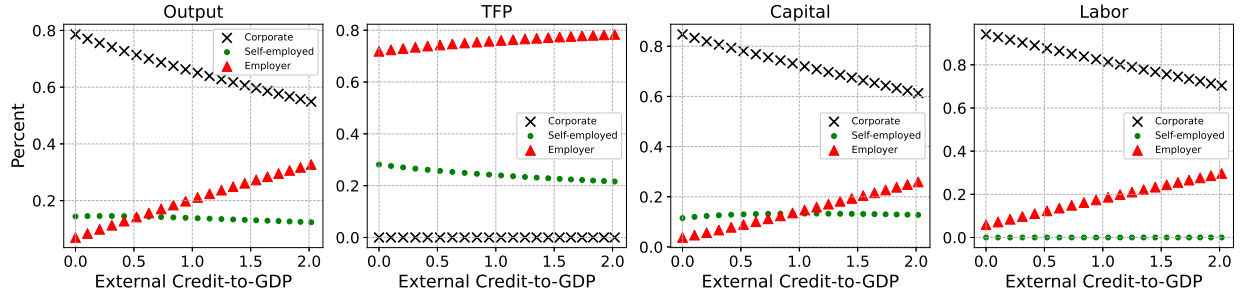


Figure 5: Long-Run Impact of Financial Development: Sectoral Shares

Note: The figure plots the steady-state shares of the corporate sector, self-employed, and employer entrepreneurs in output, TFP, capital, and labor as a function of external credit-to-GDP.

consistent with the shrinking share of workers in the occupational distribution.

TFP also rises by more than 12 percent but displays a moderate degree of curvature. This behavior reflects the way aggregate TFP is constructed as the sum of endogenous TFPs across entrepreneurial sectors. Self-employed TFP falls slightly: although misallocation within the self-employed sector improves, many relatively productive self-employed entrepreneurs exit into employer entrepreneurship, lowering the aggregate productivity of the remaining self-employed sector. On the other hand, employer TFP rises as more entrepreneurs enter and scale up. The combination generates mild curvature in overall TFP, although it continues to rise monotonically.¹⁴

Having established the aggregate steady state effects of financial development, I now decompose the output, capital, and labor into the corporate sector, self-employed entrepreneurs, and employer entrepreneurs. Figure 5 shows that the shift in shares towards the employer sector, whose share of output rises dramatically as resources are reallocated toward larger-scale entrepreneurship.

In terms of TFP, the employer sector accounts for the vast majority of measured productivity, and its share increases modestly with financial development. For factors of production, the patterns differ. Among the self-employed, even though population shares decline somewhat, their capital share remains relatively flat, and by construction they do not employ outside labor. By contrast, there is a substantial reallocation of both capital

¹⁴Appendix B presents additional steady state results on welfare, factor prices, labor market outcomes, and the TFP decomposition.

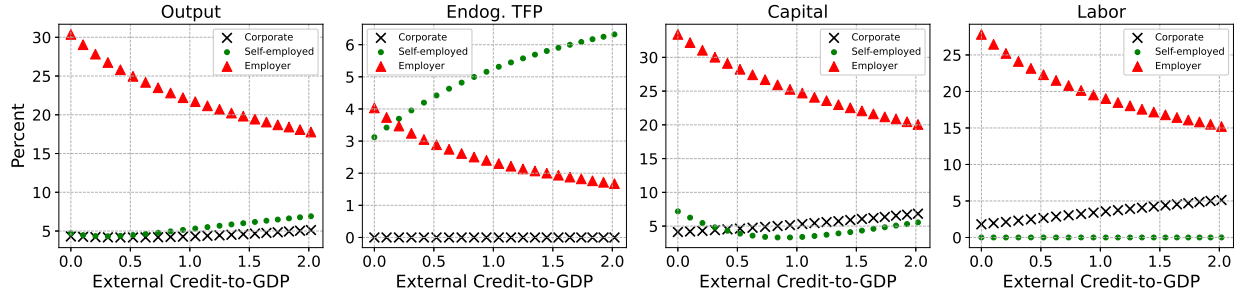


Figure 6: Impact of Financial Development: Sectoral Volatility

Note: The figure plots the standard deviations of output, factor inputs, and endogenous TFP for the corporate sector, self-employed entrepreneurs, and employer entrepreneurs. All volatilities are expressed as percentages.

and labor from the corporate sector to employers. The employer share of capital and labor rises to almost 30 percent, with corporate shares falling by comparable amounts.

This reallocation translates into a dramatic change in output composition. In a fully self-financed economy, the corporate sector produces nearly 80 percent of total output while employers account for less than 10 percent. When the external credit-to-GDP ratio reaches 2, however, the employer share rises to over 30 percent, while the corporate share falls to 55 percent. Self-employed shares decline only slightly and remain relatively stable, from 14 to 12 percent.

Overall, these results highlight the central mechanism of the model: financial development encourages a reallocation of resources from corporate firms toward employer entrepreneurship, amplifying the role of high-scale entrepreneurs in driving aggregate outcomes. This sectoral shift is one of the key channels through which improved access to credit shapes business cycle dynamics.

Having documented how financial development reallocates resources across sectors, I now turn to analyzing the second moments of the model. Figure 6 plots sectoral volatilities, measured as standard deviations, for output, factor inputs, and endogenous TFP. The figure shows that financial development not only changes where production takes place, but also changes how aggregate risk is transmitted across sectors.

I first consider endogenous TFP and factor inputs. Endogenous TFP is less volatile than output in both entrepreneurial sectors, but its response differs sharply across them.

For employer entrepreneurs, endogenous TFP volatility declines steadily as collateral constraints are relaxed. By contrast, self-employed endogenous TFP volatility rises. This increase reflects changes in selection into self-employment as financial development deepens: some relatively productive self-employed entrepreneurs transition into employer entrepreneurship, while more marginal entrepreneurs remain in or enter self-employment. As a result, the composition of the self-employed sector becomes more sensitive to aggregate conditions. The corporate sector has no endogenous TFP component by construction, so its measured endogenous TFP volatility is zero.

The factor responses reinforce the central role of employer entrepreneurship. Employer capital and labor are much more volatile than the corresponding inputs in the corporate and self-employed sectors, reflecting employers' use of hired labor and their stronger exposure to aggregate productivity and financial shocks. However, employer input volatility declines substantially with financial development. This stabilization reflects improved access to credit and greater wealth accumulation among employer entrepreneurs, which allow them to adjust production more smoothly in response to aggregate shocks. As employers rely less heavily on binding collateral constraints, their capital and labor choices become less sensitive to fluctuations in aggregate conditions. By contrast, the volatility of corporate factor inputs rises as the corporate sector absorbs more of the general equilibrium adjustment in wages and interest rates. For the self-employed, capital volatility remains relatively low and non-monotonic, while hired labor is zero by construction.

These factor inputs and endogenous TFP patterns translate into different volatility responses across sectoral outputs. Employer output remains by far the most volatile sector throughout the experiment, but its volatility declines sharply as financial development deepens, falling from about 31 percent in the self-financed economy to about 17 percent when external credit-to-GDP reaches 2. Self-employed output volatility rises moderately, from about 5 percent to about 7 percent, largely reflecting the increase in endogenous TFP volatility in the self-employed sector. Corporate output volatility increases only modestly and remains relatively low.

In sum, Figure 6 shows that financial development dampens output volatility in the

employer sector, and that this decline is quantitatively much larger than the moderate increase in self-employed output volatility. However, this sectoral stabilization does not imply a decline in aggregate risk exposure. Employers remain much more volatile than the corporate and self-employed sectors throughout the experiment, even after their volatility declines. Since financial development also reallocates output, capital, and labor toward employer entrepreneurs, the aggregate economy becomes more exposed to this shock-sensitive sector. The composition effect can therefore dominate the stabilization of employer-sector volatility, helping to explain why aggregate output volatility rises even though employer-sector volatility falls.

I quantify these forces more directly through a variance decomposition. Log-linearized aggregate output can be written as

$$\hat{Y}_t = \theta_c \hat{Y}_t^c + \theta_{se} \hat{Y}_t^{se} + \theta_{em} \hat{Y}_t^{em}, \quad (29)$$

where θ_j denotes the steady-state output share of sector $j \in c, se, em$. The variance of aggregate output is then

$$var(\hat{Y}_t) = \sum_j \theta_j^2 var(\hat{Y}_t^j) + \sum_{j \neq k} \theta_j \theta_k cov(\hat{Y}_t^j, \hat{Y}_t^k). \quad (30)$$

This expression shows that aggregate volatility depends both on within-sector volatilities and output shares across sectors. Let $var_0(\hat{Y}_t)$ denote the variance in the self-financed economy. For a given level of financial development i , the change in variance relative to self-financed economy is

$$\Delta_i var(\hat{Y}_t) = var_i(\hat{Y}_t) - var_0(\hat{Y}_t). \quad (31)$$

I decompose this change into a composition effect and a stabilization effect:

$$\Delta_i var(\hat{Y}_t) = \underbrace{(var_i(\hat{Y}_t) - var_{i'}(\hat{Y}_t))}_{\text{Composition effect}} - \underbrace{(var_{i'}(\hat{Y}_t) - var_0(\hat{Y}_t))}_{\text{Stabilization effect}}, \quad (32)$$

where $var_{i'}(\hat{Y}_t)$ is the counterfactual variance obtained by holding sectoral shares fixed

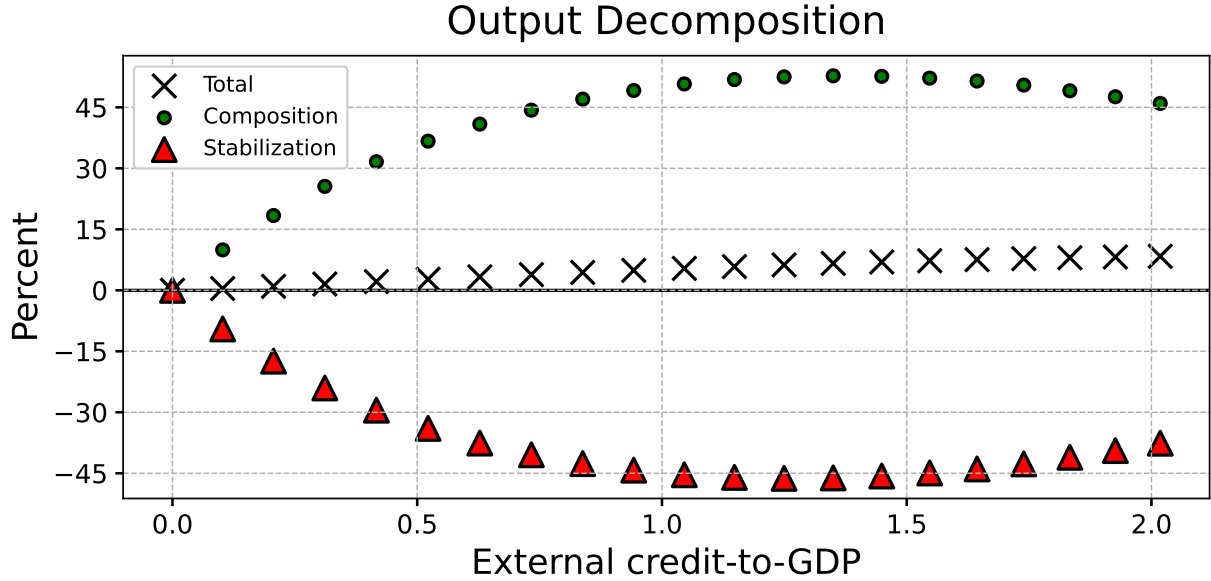


Figure 7: Impact of Financial Development: Output Volatility

Note: The figure decomposes the percentage change in the variance deviation of aggregate output relative to the self-financed economy ($\phi = 1$). Black crosses denote the total variance change, green dots the composition effect (arising from changes in steady-state output shares), and red triangles the stabilization effect (arising from changes in within-sector variances).

at the baseline while allowing only variances to adjust.¹⁵ Figure 7 plots the results. The black crosses denote the total change in aggregate output variance, while the green dots capture the composition effect and the red triangles capture the stabilization effect. The decomposition shows that financial development generates two opposing forces. The composition effect raises aggregate volatility by reallocating output toward the employer sector, which remains the most volatile sector throughout the experiment. The stabilization effect works in the opposite direction: holding sectoral output shares fixed at their baseline values, changes in sectoral second moments would reduce aggregate variance, primarily because employer sector output volatility declines substantially with financial development.

Quantitatively, the stabilization effect is large. If sectoral output shares were held fixed, the model would predict that aggregate output variance falls by as much as 46 percent. However, the composition effect is even larger, increasing aggregate variance

¹⁵This is defined as $var(\hat{Y}_t) = \sum_j (\theta_j^0)^2 var(\hat{Y}_t^j) + \sum_{j \neq k} \theta_j^0 \theta_k^0 cov(\hat{Y}_t^j, \hat{Y}_t^k)$.

by more than 52 percent at intermediate levels of financial development. When both channels operate, the composition effect dominates, and aggregate output variance rises by more than 8 percent when external credit-to-GDP reaches 2.

The figure also reveals nonlinear dynamics. At low levels of financial development, both the positive composition effect and the negative stabilization effect increase sharply in absolute value. This reflects large adjustments in both steady state sectoral shares and sectoral second moments as the economy moves away from the self-financing benchmark. As financial development deepens further, both effects flatten and eventually weaken. The stabilization effect becomes smaller in absolute value because the decline in employer sector volatility is partly offset by modest increases in volatility in the corporate and self-employed sectors. At the same time, the composition effect remains positive because employer entrepreneurs continue to account for a rising share of economic activity and remain the most volatile sector. As a result, the total effect on aggregate output variance remains positive even though the stabilizing force from employer sector volatility is quantitatively important.

This decomposition underscores a broader trade-off. On the one hand, financial development improves allocative efficiency and raises steady state output and TFP. On the other hand, it reallocates resources toward the most volatile sector, amplifying aggregate business cycle risk. To fully assess these dynamics, however, it is not enough to focus only on output volatility. I next examine how financial development shapes the volatility of consumption, investment, and prices.

Figure 8 shows how the volatility of aggregate consumption, investment, the interest rate, and wages changes as the economy's external credit-to-GDP ratio rises. Consistent with the results for output, volatility increases for most aggregate variables as financial development deepens. Consumption volatility rises by about 5 percent relative to the self-financed economy, while investment volatility increases by about 3 percent. Interest-rate volatility rises more sharply, increasing by more than 7 percentage points, reflecting the greater sensitivity of capital-market conditions as credit expands. By contrast, wage volatility remains relatively stable and follows a non-monotonic pattern. As labor demand becomes more concentrated in the employer sector, it initially amplifies wage fluc-

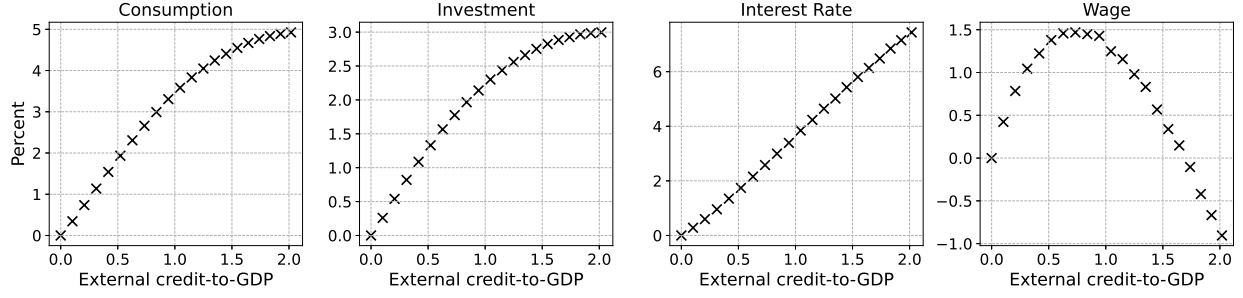


Figure 8: Impact of Financial Development: Quantities and Prices Volatility

Note: The figure plots changes in volatility relative to the self-financed economy. For consumption, investment, and wages, the y-axis reports the percent change in the standard deviation. For the interest rate, the y-axis reports the percentage-point change in its standard deviation.

tuations but later stabilizes as employers accumulate wealth and scale, making their labor demand less sensitive to aggregate shocks.

The previous decomposition separated aggregate output volatility into composition and stabilization effects, clarifying why volatility rises with financial development despite the stabilization of employer-sector output volatility. While this analysis explains the mechanism, it does not reveal which groups within the economy bear responsibility for volatility in different aggregates, which is crucial in identifying the source of aggregate risk. For instance, volatility driven primarily by employers has very different implications than volatility borne by workers, since their income sources, labor market dynamics, and exposure to financial constraints differ. To address this, I turn to a complementary decomposition that attributes the variance of output, consumption, and investment to their underlying group-level components. For any aggregate variable, $\hat{X}_t$, its variance can be expressed as a share-weighted sum of its components.

$$var(\hat{X}_t) = \sum_j \theta_j cov(\hat{X}_t, \hat{X}_t^j), \quad (33)$$

where θ_j is the steady-state share of component j , $\hat{X}_t$ denotes the log-linearized variable. Dividing both sides by the variance of $\hat{X}$, I can decompose the total variance into contri-

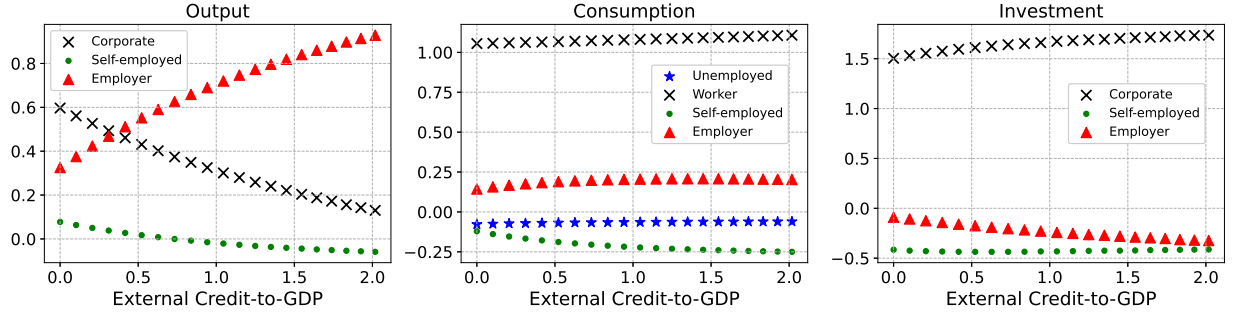


Figure 9: Short-Run Impact of Financial Development: Variance Contribution by Group

Notes: The figure plots the percentage contribution of each component to the variance of the corresponding aggregate variable as a function of the external credit-to-GDP. The components for the output and investment decompositions are the three production sectors. The components for the consumption decomposition are the four household types.

butions.

$$1 = \sum_j \theta_j \frac{cov(\hat{X}_t, \hat{X}_t^j)}{var(\hat{X}_t)} = \sum_j \theta_j \beta_j. \quad (34)$$

Here, β_j is the “beta” of the component’s volatility with respect to the aggregate’s volatility. The term, $\theta_j \beta_j$, represents the total percentage contribution of component j to the aggregate variance. I apply this decomposition to aggregate output, consumption, and investment. Figure 9 plots the resulting variance contributions. The output decomposition reinforces the main composition channel. As financial development deepens, the employer sector accounts for a steadily larger share of aggregate output volatility. Its contribution rises from around 0.3 in the self-financed economy to more than 0.9 when external credit-to-GDP reaches 2. At the same time, the corporate contribution declines sharply, while the contribution of self-employment remains small and eventually turns slightly negative. Thus, aggregate output fluctuations become increasingly tied to the employer sector as credit constraints are relaxed.

The consumption decomposition shows a different pattern. Workers remain the dominant source of aggregate consumption volatility at all levels of financial development, mainly because they account for the largest share of the population. Their contribution exceeds one because the unemployed and self-employed components contribute negatively to aggregate consumption variance. These negative contributions do not mean that

these groups have negative volatility; rather, their aggregate consumption components covary negatively with aggregate consumption. This reflects the countercyclical nature of unemployment and self-employment: during downturns, individuals are more likely to move into unemployment or necessity-driven self-employment. As financial development deepens, the negative contributions of these groups become larger in magnitude, while the employer contribution rises modestly. However, these changes remain quantitatively limited relative to the output decomposition.

Investment volatility reveals a more complex dynamic. Entrepreneurial investment contributes negatively to aggregate investment volatility. This arises because a transitory positive financial shock leads to an immediate boom in entrepreneurs' capital accumulation, followed by sharp disinvestment as forward-looking entrepreneurs anticipate tighter future credit conditions. Despite this negative correlation, the overall magnitude of employers' contribution rises with financial development, indicating that entrepreneurial investment decisions become increasingly important for aggregate fluctuations.

In summary, this section highlights that looser credit constraints shift the economy toward employer entrepreneurship. Because this sector is more exposed to aggregate shocks, the result is heightened macroeconomic volatility.

6 Conclusion

This paper examines how entrepreneurial heterogeneity shapes macroeconomic dynamics. Using microdata from Argentina, I document two key facts. First, entrepreneurial types differ systematically across the income distribution: self-employed entrepreneurs are disproportionately concentrated below the median, while employer entrepreneurs are concentrated above the median. Second, these groups move differently over the business cycle. The population share of self-employed entrepreneurs is countercyclical, while the employer share is procyclical.

Motivated by these patterns, I develop a dynamic general equilibrium model with occupational choice, financial and labor market frictions, and aggregate shocks. The model

replicates the distributional and cyclical features of entrepreneurship observed in the data and captures endogenous sorting across occupations by wealth and ability. Quantitative experiments show that financial development raises long-run output but also increases aggregate volatility. The mechanism operates through occupational composition. Relaxing collateral constraints shifts economic activity toward employer entrepreneurs, who operate larger firms and remain more exposed to aggregate shocks. Although financial development dampens output volatility within the employer sector, employers remain the most volatile sector, and their rising aggregate weight increases the economy's exposure to business cycle risk.

These findings underscore a trade-off between long-run growth and short-run stability. Policies that expand access to credit delivers higher output in the long-run, but they can also amplify business cycle fluctuations unless accompanied by complementary structural reforms that dampen volatility. More broadly, the analysis highlights the importance of explicitly accounting for heterogeneity within the entrepreneurial sector when evaluating the macroeconomic consequences of financial development.

References

- Aguiar, M. and Gopinath, G. (2007). Emerging Market Business Cycles: The Cycle Is the Trend. *Journal of Political Economy*, 115(1):69–102.
- Allub, L. and Erosa, A. (2019). Financial frictions, occupational choice and economic inequality. *Journal of Monetary Economics*, 107:63–76.
- Alvaredo, F. (2010). The rich in Argentina over the twentieth century, 1932–2004. In Atkinson, A. B. and Piketty, T., editors, *Top Incomes over the Twentieth Century: A Global Perspective*, chapter 6, pages 253–298. Oxford University Press, Oxford.
- Bayer, C. and Luetticke, R. (2020). Solving heterogeneous agent models in discrete time with many idiosyncratic states by perturbation methods. *Quantitative Economics*, 11:1253–1288.
- Bosch, M. and Maloney, W. F. (2010). Comparative analysis of labor market dynamics using markov processes: An application to informality. *Labour Economics*.
- Buera, F. J., Kaboski, J. P., and Shin, Y. (2011). Finance and Development: A Tale of Two Sectors. *American Economic Review*, 101(5):1964–2002.
- Buera, F. J., Kaboski, J. P., and Shin, Y. (2015). Entrepreneurship and Financial Frictions: A Macro-Development Perspective. *Annual Review of Economics*, page 39.
- Buera, F. J., Kaboski, J. P., and Shin, Y. (2021). The Macroeconomics of Microfinance. *Review of Economic Studies*, page 40.
- Cagetti, M. and De Nardi, M. (2006). Entrepreneurship, Frictions, and Wealth. *Journal of Political Economy*, 114(5):36.
- Cerutti, E., Claessens, S., and Laeven, L. (2017). The use and effectiveness of macroprudential policies: New evidence. *Journal of Financial Stability*, 28:203–224.
- Denizer, C., Iyigun, M., and Owen, A. (2002). Finance and macroeconomic volatility. *The B.E. Journal of Macroeconomics*.
- Donovan, K., Lu, W. J., and Schoellman, T. (2023). Labor market dynamics and development*. *The Quarterly Journal of Economics*, 138(4):2287–2325.
- Evans, D. S. and Jovanovic, B. (1989). An Estimated Model of Entrepreneurial Choice under Liquidity Constraints. *Journal of Political Economy*, 97(4):808–827.

- Feng, Y. and Ren, J. (2023). Skill bias, financial frictions, and selection into entrepreneurship. *Journal of Development Economics*, 162:103046.
- Fernández, A. and Meza, F. (2015). Informal employment and business cycles in emerging economies: The case of Mexico. *Review of Economic Dynamics*, 18(2):381–405.
- Finamor, L. (2025). Labor market informality, risk, and insurance. *mimeo*.
- Finkelstein–Shapiro, A. (2014). Self-employment and business cycle persistence: Does the composition of employment matter for economic recoveries? *Journal of Economic Dynamics and Control*, 46:200–218.
- Genz, A. (1992). Numerical computation of multivariate normal probabilities. *Journal of Computational and Graphical Statistics*, 1(2):141–149.
- Guner, N., Ventura, G., and Xu, Y. (2008). Macroeconomic implications of size-dependent policies. *Review of Economic Dynamics*, 11(4):721–744.
- Herreño, J. and Ocampo, S. (2023). The macroeconomic consequences of subsistence self-employment. *Journal of Monetary Economics*, pages 91–106.
- Hong, S. (2023). Emerging Market Business Cycles with Heterogeneous Agents. *mimeo*.
- Horvath, J. (2018). Business cycles, informal economy, and interest rates in emerging countries. *Journal of Macroeconomics*, 55:96–116.
- Hsieh, C.-T. and Klenow, P. J. (2009). Misallocation and manufacturing tfp in china and india. *The Quarterly Journal of Economics*, 124(4):1403–1448.
- Jaar, D. (2023). Self-employment as self-insurance. *mimeo*.
- Jermann, U. and Quadrini, V. (2012). Macroeconomic Effects of Financial Shocks. *American Economic Review*, 102(1):238–271.
- Kiyotaki, N. and Moore, J. (1997). Credit cycles. *Journal of Political Economy*, 105(2):211–248.
- Krusell, P. and Smith, Jr., A. A. (1998). Income and Wealth Heterogeneity in the Macroeconomy. *Journal of Political Economy*, 106(5):867–896.
- Kwark, N.-S. and Ma, E. (2020). Entrepreneurship and Income Distribution Dynamics: Why Is the Income Share of Top Income Earners Acyclical over the Business Cycle? *International Economic Review*, page 53.
- Loayza, N. V. and Rigolini, J. (2011). Informal employment: Safety net or growth engine?

- World Development*, 39(9):1503–1515.
- Lucas, R. E. (1978). On the Size Distribution of Business Firms. *The Bell Journal of Economics*, 9(2):508–523.
- McKiernan, K. (2021). Social security reform in the presence of informality. *Review of Economic Dynamics*, 40:228–251.
- Midrigan, V. and Xu, D. Y. (2014). Finance and Misallocation: Evidence from Plant-Level Data. *American Economic Review*, 104(2):37.
- Poschke, M. (2012). Who becomes an entrepreneur? labor market prospects and occupational choice. *Journal of Economic Dynamics and Control*.
- Quadrini, V. (2000). Entrepreneurship, Saving, and Social Mobility'. *Review of Economic Dynamics*, page 40.
- Ranciere, R., Tornell, A., and Westermann, F. (2006). Decomposing the effects of financial liberalization: Crises vs. growth. *Journal of Banking Finance*, 30(12):3331–3348.
- Reinhart, C. M. and Rogoff, K. S. (2009). *This Time is Different: Eight Centuries of Financial Folly*. Princeton University Press, Princeton, New Jersey.
- Reiter, M. (2009). Solving heterogeneous-agent models by projection and perturbation. *Journal of Economic Dynamics and Control*, 33(3):649–665.
- Restrepo-Echavarria, P. (2014). Macroeconomic volatility: The role of the informal economy. *European Economic Review*, 70:454–469.
- Storesletten, K., Telmer, C. I., and Yaron, A. (2004). Consumption and risk sharing over the life cycle. *Journal of Monetary Economics*, 51(3):609–633.
- Young, E. R. (2010). Solving the incomplete markets model with aggregate uncertainty using the Krusell–Smith algorithm and non-stochastic simulations. *Journal of Economic Dynamics and Control*, 34(1):36–41.

A Empirical Appendix

This appendix provides additional details on the construction of the empirical variables used in Section 2.

A.1 Construction of Income Variable

Total individual income is measured using *P47T*. This variable is broader than income from the respondent's primary occupation alone. It aggregates labor and non-labor income received by the individual during the reference month. Labor income includes income from the main occupation, with separate reporting depending on whether the respondent is a wage or salaried employee, self-employed worker, or employer. For wage and salaried workers, the components include wages and salaries, commissions from sales or production, tips, extraordinary or irregular bonuses, and retroactive payments. For self-employed workers and employers, the relevant components measure business or professional income received by the respondent, with separate reporting depending on whether the activity is operated individually or with partners or family members.

Non-labor income includes unemployment insurance, retirement income and pensions, government or private transfers and social assistance, rental income, profits from businesses in which the individual does not actively work, interest and investment income, scholarships, alimony and other income support, other cash income, and retirement or pension bonuses.

A.2 Measurement issues

Several measurement issues are important for interpreting the empirical results. First, because *P47T* is collected in a household survey, it is self-reported and cannot be cleanly interpreted as either pre-tax or post-tax income. Second, survey income is subject to underreporting and undercoverage, especially at the top of the income distribution. This issue is particularly relevant for capital income and for the top 0.1 and 0.01 percent of the distribution. For Argentina, Alvaredo (2010) documents substantial discrepancies be-

tween household survey data and tax-return data at the very top. Third, the EPH does not provide data on wealth, which limits the paper’s empirical analysis on wealth comparisons across self-employed and employer entrepreneurs.

At the same time, household survey data have advantages for the purposes of this paper. In particular, the EPH provides broad coverage of labor market activity, including individuals operating outside the formal sector, which is especially important for measuring self-employment. Moreover, the empirical motivation in the paper is based on broad occupational categories and income groups rather than top-income shares or wealth. For this reason, while income underreporting and undercoverage as well as wealth not being measured are important limitations, they are less central here than they would be in a study focused on the very top of the income distribution.

A.3 Calibration of the Worker Productivity Process

This appendix describes the empirical procedure used to discipline the worker productivity process. In the model, worker productivity satisfies

$$x_{i,t+1} = \rho_w x_{it} + \varepsilon_{i,t+1}, \quad \varepsilon_{i,t+1} \sim N(0, \sigma_w^2), \quad (\text{A1})$$

where the model period is one quarter. The objective of the exercise is to calibrate the persistence parameter ρ_w and the innovation standard deviation σ_w using the short panel dimension of the EPH. The procedure follows the covariance-based approach commonly used in the earnings-dynamics literature; see, for example, Storesletten et al. (2004).

The empirical measure of worker earnings is real labor income from the main occupation. I restrict the sample to employed wage workers with positive main occupation earnings. I first remove predictable earnings differences by estimating

$$\log y_{it} = \alpha + \delta_{\text{age}(it)} + \delta_{\text{sex}(i)} + \delta_{\text{educ}(it)} + \delta_{\text{loc}(it)} + \lambda_t + r_{it}, \quad (\text{A2})$$

where λ_t denotes calendar-quarter fixed effects. The age, sex, education, location, and calendar-quarter variables are all treated as categorical variables. I do not include individual fixed effects, since doing so would remove much of the low-frequency individual

component that the calibration is meant to measure. The residual r_{it} is the earnings object used in the second step.

In the second step, I exploit the fact that EPH is an unbalanced short panel. Since many individuals are observed only once, and repeated observations are not always consecutive, I use all pairs of observations for the same worker. For each valid pair, I compute

$$\Delta r_{i,t,t'} = r_{it'} - r_{it}, \quad (\text{A3})$$

and then construct the empirical earnings-change moment

$$\hat{D}_h = \frac{1}{N_h} \sum_{(i,t,t'): q(t')-q(t)=h} (r_{it'} - r_{it})^2, \quad (\text{A4})$$

where $h > 0$. In the baseline sample, the available lags are $h \in \{1, 3, 4, 5\}$.

To map these moments into the model parameters, I write residual earnings as

$$r_{it} = a_i + x_{it} + u_{it}, \quad (\text{A5})$$

where a_i is a permanent individual component, x_{it} is the persistent AR(1) component, and u_{it} is transitory earnings variation or measurement error. The transitory component is included only for identification of the earnings moments; it is not added to the structural model. Differencing removes a_i . Under stationarity of x_{it} ,

$$\text{Var}(x_{it}) = \frac{\sigma_w^2}{1 - \rho_w^2}. \quad (\text{A6})$$

Therefore, we can write the moment conditions of differences in residual earnings as

$$D_h = E \left[(r_{i,t+h} - r_{it})^2 \right] = 2\sigma_w^2 \frac{1 - \rho_w^h}{1 - \rho_w^2} + 2\sigma_u^2, \quad (\text{A7})$$

where $\sigma_u^2 = \text{Var}(u_{it})$. I estimate $\theta = (\rho_w, \sigma_w^2, \sigma_u^2)$ by minimum distance:

$$\hat{\theta} = \arg \min_{\rho_w, \sigma_w^2, \sigma_u^2} \sum_{h \in \{1, 3, 4, 5\}} \left[\hat{D}_h - \left(2\sigma_w^2 \frac{1 - \rho_w^h}{1 - \rho_w^2} + 2\sigma_u^2 \right) \right]^2. \quad (\text{A8})$$

Because the panel covers only a short horizon, the objective is flat near high persistence values. For this reason, I use the EPH moments primarily to discipline the persistent inno-

Table 10: Worker income moments

Horizon h	1	3	4	5
$\hat{D}_h^{\text{Data}}$	0.267	0.293	0.310	0.325
$\hat{D}_h^{\text{Model}}$	0.265	0.296	0.310	0.324

vation variance conditional on a high quarterly persistence. I set the quarterly persistence to 0.96 to be consistent with the value used in Allub and Erosa (2019). The estimated moments are With quarterly persistence of $\rho_w = 0.96$, the minimum-distance estimate gives $\hat{\sigma}_w = 0.126$ and $\hat{\sigma}_u = 0.353$.

B Additional Steady State Model Results

B.1 Welfare

I examine how aggregate welfare changes in the steady state. Welfare is measured in consumption-equivalent units, defined as the permanent percentage increase in consumption that makes households indifferent between the self-financed economy (with subscript “0”) and alternative economies (denoted with tildes) with greater credit access. Formally, the aggregate consumption-equivalent welfare measure, λ , is given by

$$\lambda = \left(\frac{\int \tilde{v}_i \tilde{d}_i}{\int v_{0i} d_{0i}} \right)^{\frac{1}{1-\gamma}} - 1, \quad (\text{A9})$$

where γ is the coefficient of relative risk aversion.

Figure 10 reports the results. Consistent with prior work on financial development, welfare gains are positive and sizable. Greater access to credit allows factors to reallocate toward more productive entrepreneurs, raising efficiency, output, and capital accumulation. These gains more than offset the decline in aggregate labor supply as individuals exit wage employment to pursue employer entrepreneurship. Quantitatively, when the external credit-to-GDP ratio rises above 2, aggregate welfare increases by roughly 4 percent in consumption-equivalent terms. It must be noted that these welfare gains do not provide a complete picture. Because the experiment only compares steady states, it abstracts

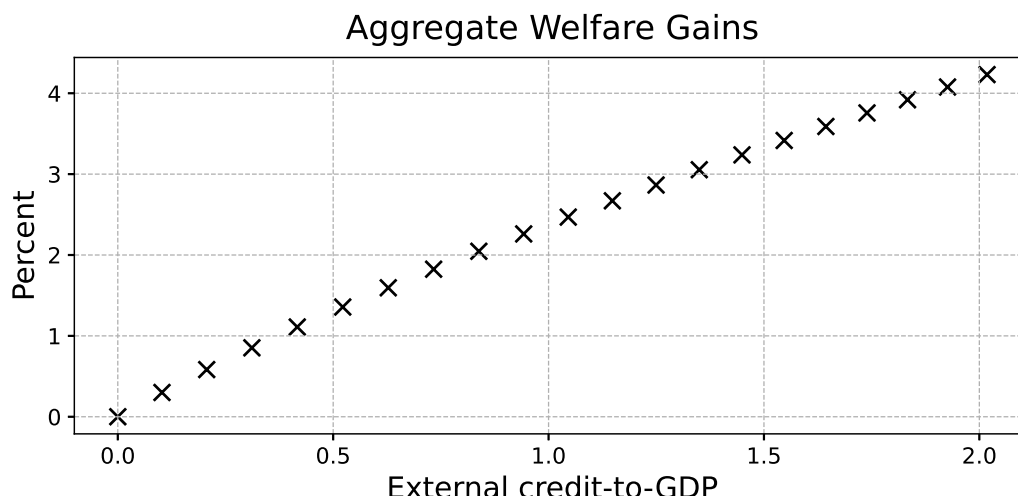


Figure 10: Steady-State Welfare Gains from Financial Development

from the transitional dynamics associated with capital accumulation and occupational reallocation. Accordingly, the reported gains should be interpreted as comparisons across stationary equilibria rather than as a full welfare evaluation of a reform; incorporating the transition path could yield different welfare effects.

B.2 Factor Prices and the Labor Market

This subsection further examines the labor market mechanism behind the decline in the worker population share and market labor input when collateral constraints are relaxed. The key point is that this decline does not reflect a decline in employer labor demand. Instead, it reflects a general equilibrium reallocation of individuals and labor demand across occupations and production sectors. Figure 11 plots the wage and interest rates in the steady state as the external credit-to-GDP ratio rises. Financial development raises aggregate productivity and increases capital accumulation, both of which affect factor prices. The wage rate rises by 1.7 percent, reflecting the higher marginal product of labor. By contrast, the interest rate falls by about 40 basis points in annualized terms, as the expansion in aggregate capital makes capital less scarce in equilibrium.

Next, Figure 12 plots changes in market labor demand. Employer labor demand rises monotonically with financial development, reflecting both entry into employer en-

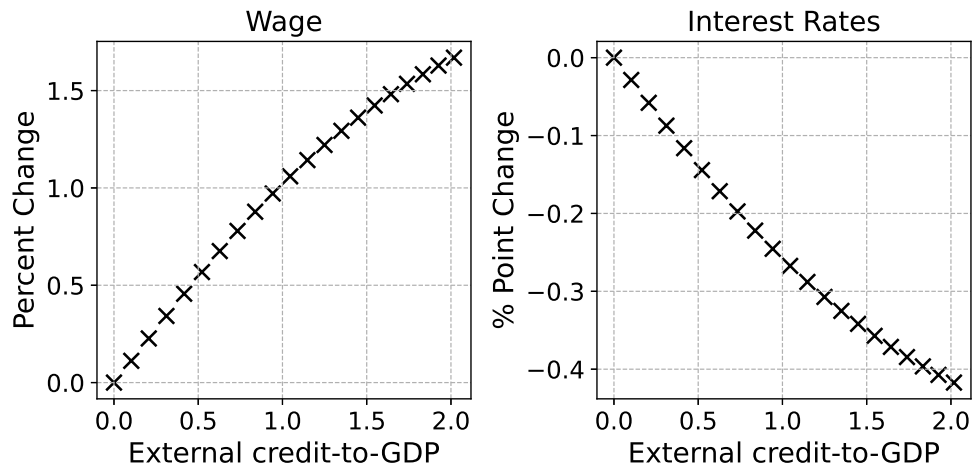


Figure 11: Steady-State Factor Prices under Financial Development

Note: The figure plots changes in steady-state factor prices relative to the self-financed economy. The wage panel reports percent changes, while the interest-rate panel reports annualized percentage point changes.

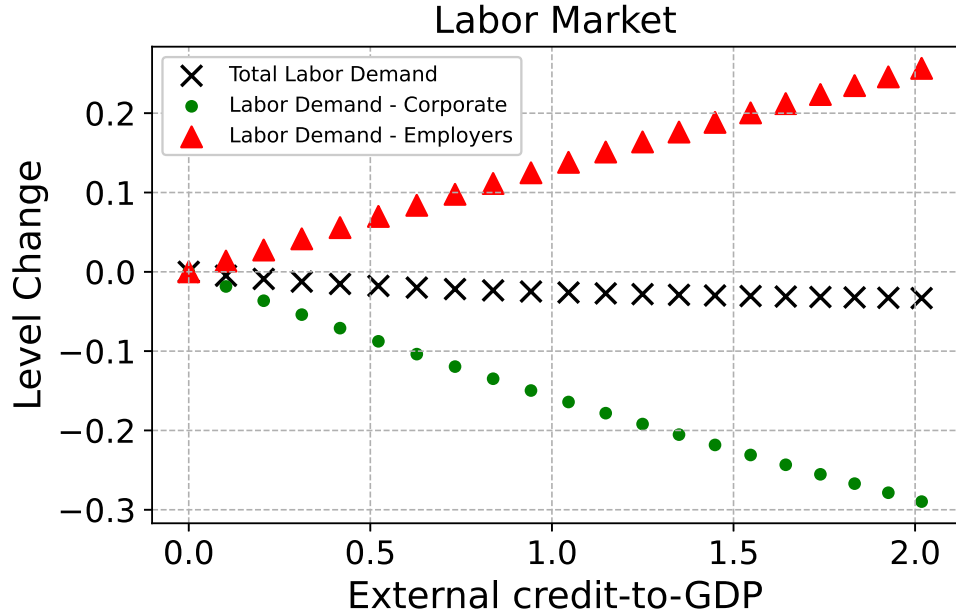


Figure 12: Labor Demand under Financial Development

Note: The figure plots level changes relative to the self-financed economy. Total labor demand is decomposed into corporate sector demand and employer entrepreneur demand.

trepreneurship and the expansion of existing employer firms. At the same time, corporate labor demand declines as resources are reallocated away from the corporate sector and toward employer entrepreneurs. The decline in corporate labor demand is slightly larger than the increase in employer labor demand, so total effective market labor declines modestly in equilibrium.

This pattern explains why market labor can fall even though employer entrepreneurship expands. Relaxing collateral constraints raises the return to employer entrepreneurship relative to wage work. As a result, some individuals who would otherwise supply labor in the wage market instead become employer entrepreneurs and devote their time to management. Employer labor demand therefore rises, but the mass of wage workers declines. In equilibrium, the increase in employer labor demand is more than offset by the decline in corporate labor demand and therefore leads to a reduction in total market labor demand.

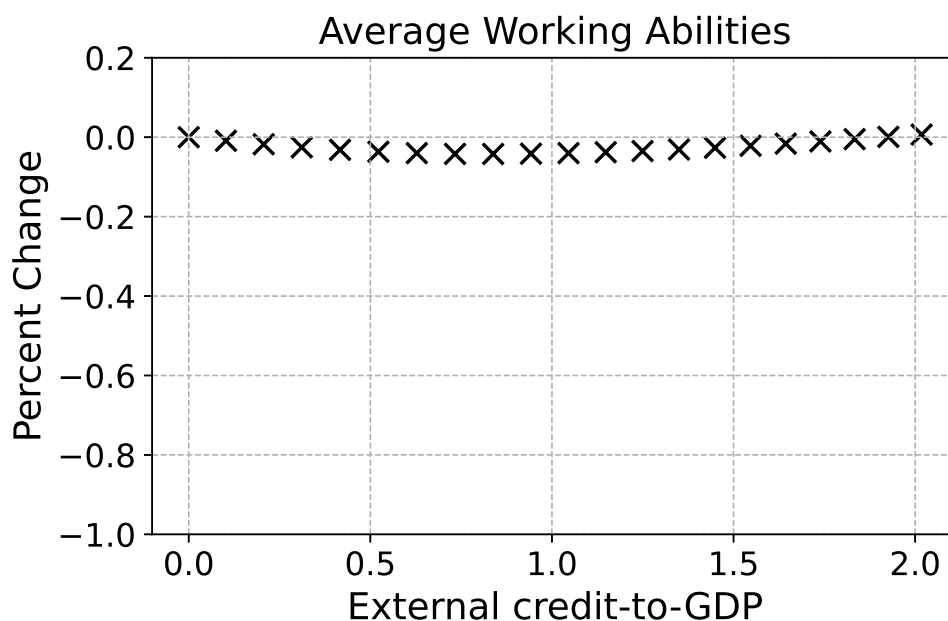


Figure 13: Average Working Ability among Wage Workers

Note: The figure plots the percent change in average worker ability among households who choose wage work, relative to the self-financed economy.

Finally, I examine whether the decline in market labor is driven only by the extensive

margin, or also by changes in the composition of remaining wage workers. Figure 13 reports the average worker ability among households who remain in wage employment. The figure shows that changes in the average ability of wage workers are quantitatively small. Over the full range of financial development, average worker ability changes by less than 0.05 percent. Thus, the decline in market labor is driven primarily by the decline in the number of wage workers, rather than by a deterioration in the ability composition of workers.

The figure displays a mild non-monotonic pattern. At low levels of financial development, average worker ability among wage workers falls slightly. This reflects selection into entrepreneurship. Since worker ability and managerial ability are positively correlated, individuals with high managerial ability also tend to have relatively high worker ability. When credit constraints first loosen, some of these individuals leave wage work to enter entrepreneurship, which lowers the average worker ability of those who remain in wage employment.

At higher levels of financial development, this pattern reverses. As wages rise, wage work becomes increasingly attractive for households with high worker ability. At the same time, deeper financial development allows entrepreneurship to expand among individuals with high managerial ability relative to their worker ability. These selection forces raise the average worker ability of the remaining wage workers. However, the magnitude of this compositional change is very small. Therefore, the main force behind the decline in market labor is the extensive margin of individuals out of wage work and into entrepreneurship, especially employer entrepreneurship.

B.3 TFP Decomposition of Entrepreneurs

I next examine the forces driving the response of endogenous TFP to financial development. Let n_t^j denote the population mass of entrepreneurial type $j \in \{se, em\}$. The TFP measure can then be written as

$$TFP = (n^{se})^{1-\eta} \bar{A}^{se} + (n^{em})^{1-\eta-\theta} \bar{A}^{em}, \quad (\text{A10})$$

where $\bar{A}^{se}$ and $\bar{A}^{em}$ are defined as

$$\bar{A}^{se} = \frac{\bar{Y}^{se}}{(\bar{K}^{se})^\eta}, \quad \bar{A}^{em} = \frac{\bar{Y}^{em}}{(\bar{K}^{em})^\eta (\bar{L}^{em})^\theta}. \quad (\text{A11})$$

where $\bar{X}$ denote average of X or X divided by its population share. The change in aggregate TFP can be written exactly as

$$\Delta TFP = \underbrace{\sum_{j \in \{se, em\}} (n_0^j)^{\gamma_j} \Delta \bar{A}^j}_{\text{within effect}} + \underbrace{\sum_{j \in \{se, em\}} \bar{A}_0^j \Delta [(n^j)^{\gamma_j}]}_{\text{across effect}} + \underbrace{\sum_{j \in \{se, em\}} \Delta [(n_0^j)^{\gamma_j}] \Delta \bar{A}^j}_{\text{interaction effect}}, \quad (\text{A12})$$

where $\gamma_{se} = 1 - \eta$ and $\gamma_{em} = 1 - \eta - \theta$. The first term is the within effect. It captures how entrepreneurial TFP would change if group-level average TFP changed while entrepreneurial population shares remained fixed at their baseline values. The second term is the across, or composition, effect. It captures how entrepreneurial TFP would change if population shares changed while group-level average TFP remained fixed at baseline values. The third term is the interaction effect, which captures the fact that group-level average TFP and population shares change simultaneously.

Figure 14 quantifies these three components in the baseline calibration. Each component is expressed as a percentage of entrepreneurial TFP in the completely self-financed economy.

The decomposition shows that most of the increase in entrepreneurial TFP is driven by the across effect. As financial constraints are relaxed, the entrepreneurial population shifts away from self-employment and toward employer entrepreneurship. Since employer entrepreneurs have higher baseline TFP, this reallocation raises aggregate entrepreneurial TFP. By contrast, the within component is negative: after removing the effect of changes in entrepreneurial masses, the normalized productivity residuals decline modestly. This pattern is consistent with the entry of more marginal entrepreneurs and changes in factor allocation within entrepreneurial types. Overall, the positive across effect more than offsets this negative within contribution.

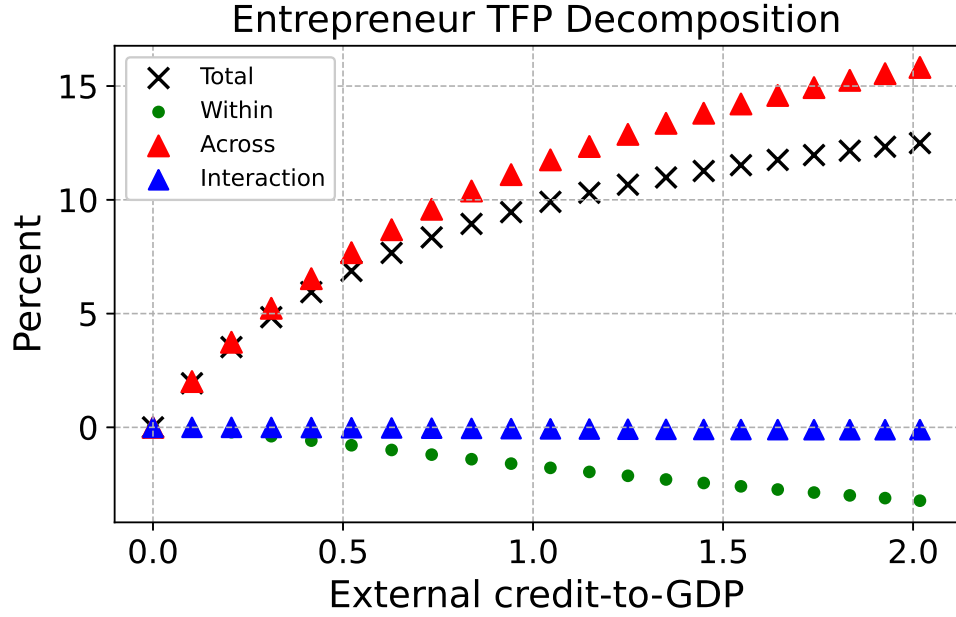


Figure 14: TFP Decomposition

C Computational Appendix

The baseline model contains multiple idiosyncratic productivity shocks, an aggregate shock, and discrete choice at the household level. The general steps to solve the model is as follows:

1. Solve for the stationary equilibrium.
 - (a) Guess the level of interest rate, r_{ss} .
 - (b) Find factor prices (U , W) from corporate first-order conditions.
 - (c) Given prices, solve the household problem, which is solved using a value function iteration with a cubic spline to evaluate the value function off the grid points. The asset grid is set from 0.05 to 300. The number of asset grid points is 200. Increasing the number of grid points did not substantially change the results. In order to capture the high degree of nonlinearity at the lower end of the grid, I include more grid points towards the bottom of grid. There are 7 grid points in the managerial productivity shock, another 7 grid points for

worker productivity, and 4 occupations. In total, there are 39,200 individual state space.

- (d) To discretize the idiosyncratic shocks, I construct a tensor grid for (x_w, x_m) , with bounds set at ± 3 unconditional standard deviations of each process. The transition probabilities are then computed directly from the correlated bivariate normal distribution, rather than as the product of two independent transition matrices. The bivariate-normal rectangle probabilities are evaluated using the quasi-Monte Carlo method of Genz (1992). This multivariate Tauchen-style discretization directly incorporates the covariance between the innovations: the transition matrix is not constructed as the product of independent marginal transition matrices, nor does the procedure require orthogonalizing the shocks through a Cholesky decomposition.
 - (e) Given the solution to the household problem, simulate the distribution as in Young (2010).
 - (f) Check that the capital market clears.
2. Linearize the model with only respect to aggregate shocks
- (a) Jacobian is calculated by taking automatic differentiations
 - (b) Use the dimension reduction techniques from Bayer and Luetticke (2020) to make the model computationally feasible. Using this technique reduces the number of state variables from 39,204 (39,200 individual states plus two aggregate shocks, aggregate capital, and lagged aggregate output) to 257 (200 asset grids plus 49 productivity grids plus 4 states for copulas plus two aggregate shocks plus aggregate capital plus lagged output). The number of control variables decline from 39,255 to 85.
3. Solve the linearized model using Klein's method.

C.0.1 Details on Solving the Model

There are two infinite dimensional objects in the model that need to be approximated, which are the value functions and the distribution of households over the idiosyncratic states. Value functions are approximated by a cubic spline. I first calculate the prices in the steady state and turn off all aggregate shocks. In this step, the conditional expectations for the value functions are only taken with respect to idiosyncratic shocks. Given that the value function can be approximated with a cubic spline, one can maximize each value functions on the RHS using Brent's method. And then the household's equilibrium dynamics with aggregate shocks can be characterized by the above set of equations, where $s = (a, z_w, z_m, o)$ are idiosyncratic state variables while $S = (Z, \phi, \Lambda^-, Y_{-1})$ will be aggregate state variables.

The high dimensionality of this system makes the computation nearly infeasible. Therefore, I pursue a dimension reduction technique as Bayer and Luetticke (2020) for the distribution of households. To reduce the dimension of the value function, compression algorithm is used. To be more specific, I write the value function as some form of sparse polynomial expansions around its stationary equilibrium values.

$$v_t(s) = \bar{v}(s) + g_v(s; \theta_v^s), \quad (\text{A13})$$

where g_v is the discrete cosine transformation of the stationary equilibrium of the value function. I shrink all but the largest elements without losing too much information. That is, I only keep the nodes of the value function where it is most informative in response to aggregate shocks. For more technical details, I refer the readers to Bayer and Luetticke (2020).

The second infinite-dimensional object is the distribution of households over idiosyncratic states. I approximate this distribution using a non-stochastic histogram method following Young (2010). Because occupational choice occurs at the beginning of the period, I distinguish between the pre-choice distribution and the post-choice distribution. Let Λ_t^- denote the beginning-of-period distribution over $i = (a, z_w, z_m, o)$ where o is the occupation chosen in the previous period.

After households observe the aggregate state and labor market shocks, they choose their current occupation $j \in \{u, w, se, em\}$. Let $\chi_j(i, S_t)$ denote the probability that a household of type i chooses current occupation j , incorporating both labor market transition probabilities and the Type-I Extreme Value choice shocks. The current occupational choices induce post-choice, occupation-specific distributions:

$$\Lambda_t^j(a, z_w, z_m) = \sum_o \chi_j(a, z_w, z_m, o; S_t) \Lambda_t^-(a, z_w, z_m, o). \quad (\text{A14})$$

These post-choice distributions are used to compute current production, factor demands, income, and market-clearing conditions. They are not additional state variables; they are auxiliary objects constructed from the pre-choice distribution and the occupational choice probabilities.

Then, we can find the distribution of households over the next period state using the savings decision as well as Markov probability matrix of productivity processes. Given current occupation j , let $a'^j(i, S_t)$ be the savings policy. The current occupation becomes next period's lagged occupation. Let $\omega_{\ell n}^j(i, S_t)$ denote the interpolation weight that assigns savings $a'^j(i, S_t)$ from current asset grid point a_ℓ to next-period asset grid point a_n , and let $\Pi_z(z'_w, z'_m | z_w, z_m)$ denote the joint transition probability for idiosyncratic skills. For $i = (a_\ell, z_w, z_m, o)$ and $i' = (a_n, z'_w, z'_m, o')$, define the transition matrix $Q(S_t)$ by

$$Q_{i', i}(S_t) = \sum_{j \in \{u, w, se, em\}} \chi_j(i, S_t) \mathbf{1}\{o' = j\} \omega_{\ell n}^j(i, S_t) \Pi_z(z'_w, z'_m | z_w, z_m). \quad (\text{A15})$$

The next-period pre-choice distribution is then

$$\Lambda_{t+1}^- = Q(S_t) \Lambda_t^-. \quad (\text{A16})$$

In the stationary equilibrium, the aggregate state is constant and the transition matrix is evaluated at the steady state, $\bar{Q} = Q(\bar{S})$. The stationary pre-choice distribution satisfies

$$\bar{\Lambda}^- = \bar{Q} \bar{\Lambda}^-. \quad (\text{A17})$$

With aggregate shocks, $Q(S_t)$ varies with the aggregate state through prices, policy func-

tions, and occupational choice probabilities, so the distribution evolves according to

$$\Lambda_{t+1}^- = Q(S_t)\Lambda_t^-. \quad (\text{A18})$$

Furthermore, there needs to be equations that describe the aggregate capital stock, and aggregate shocks. They are

$$\int a_i d\Lambda = K, \quad (\text{A19})$$

$$\int z_{wi} d\Lambda = L^c + L^{em} \quad (\text{A20})$$

$$\log Z' = (1 - \rho_z) \log \bar{Z} + \rho_z \log Z + \varepsilon'_z, \quad (\text{A21})$$

$$\log \phi' = (1 - \rho_\phi) \log \bar{\phi} + \rho_\phi \log \phi + \varepsilon'_\phi, \quad (\text{A22})$$

$$Y^c = Z(K^c)^\alpha (L^c)^\alpha, \quad (\text{A23})$$

$$Y = Y^c + \int y(s) d\Lambda_{em,se}(s), \quad (\text{A24})$$

$$W = Z(1 - \alpha)(K^c / L^c)^\alpha, \quad (\text{A25})$$

$$U = Z\alpha(K^c / L^c)^{(1-\alpha)}, \quad (\text{A26})$$

$$I = K' - (1 - \delta)K, \quad (\text{A27})$$

$$C = \int c(s) d\Lambda, \quad (\text{A28})$$

$$K = K^c + \int k(s) d\Lambda_{se,em}, \quad (\text{A29})$$

$$\int z_w(s) d\Lambda_w = L^c + \int n_d(s) d\Lambda_{em}, \quad (\text{A30})$$

This completes the minimum number of equations in order to fully characterize the equilibrium dynamics in my model.

Given the distribution and the value functions, all other auxiliary aggregate variables can be calculated from the value functions (and its resulting savings/consumption functions) and the resulting distribution.

The equilibrium dynamic can be represented by a set of nonlinear equations (shown above) which then can be written as:

$$\mathbb{E}_t F(X, X', Y, Y') = 0, \quad (\text{A31})$$

where Y is the set of control variables (such as value functions or aggregate output), and X is the set of state variables (such as the distribution Λ). Thus, linearizing the model with respect to aggregate shocks gives the following linear dynamic system.

$$X' = H_x X + \eta \epsilon', \quad (\text{A32})$$

$$Y = G_x X, \quad (\text{A33})$$

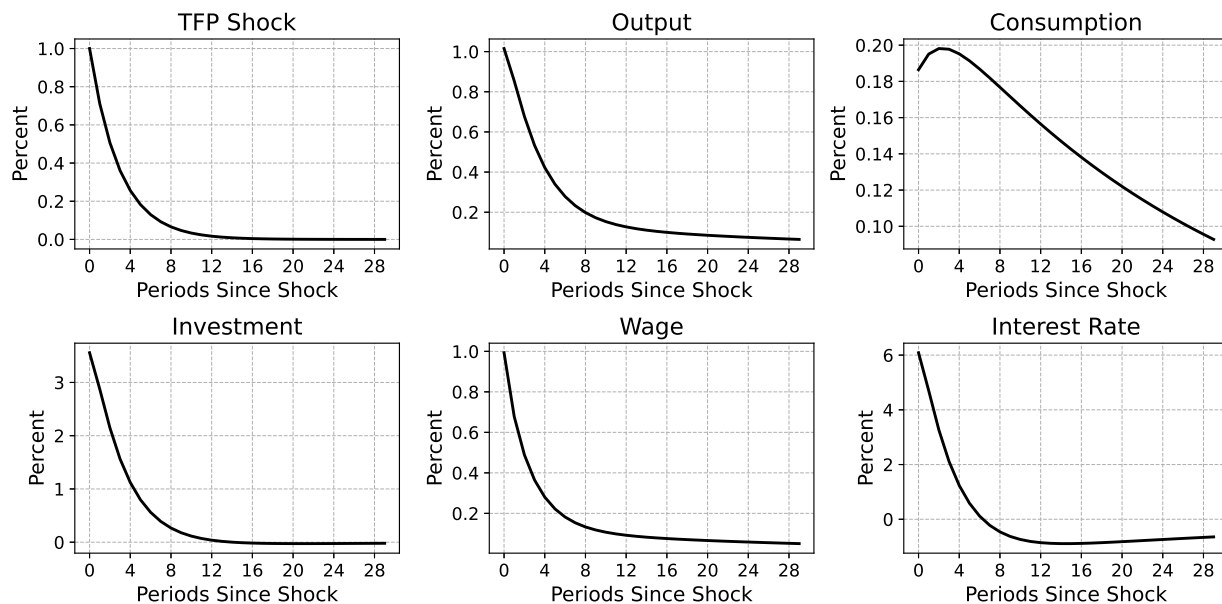
which then can simulate the model, calculate second moments, and perform impulse response analysis.

D Additional Impulse Response

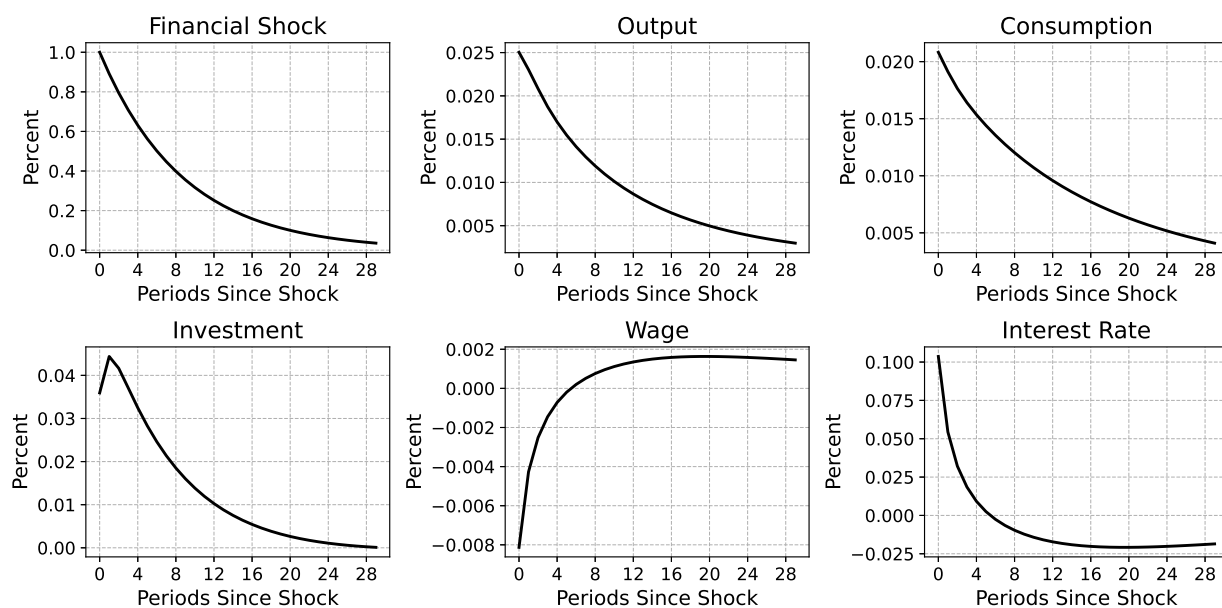
In response to TFP shocks, output, consumption, investment, wages, and interest rates all respond similarly to the representative-agent economy. Output increases by 1%, consumption responds much less than output, while investment responds more than three times as much as output. Both wages and interest rates rise following productivity shocks.

For financial shocks, output, consumption, and investment all increase on impact, though the aggregate effects are minimal because these shocks directly affect only a small share of the population. Since financial shocks typically impact entrepreneurial capital demand, they reallocate resources from the corporate sector to entrepreneurs. This reallocation ultimately leads to a rise in interest rates and a slight decline in wages.

Figure 16 plots population shares and aggregate incomes for each occupation in response to the two aggregate shocks, shown in separate panels. Following a positive TFP shock, the population shares of both workers and employers increase after 4 quarters. The worker share rises because higher wages make wage employment more attractive, while the employer share rises because higher productivity increases entrepreneurial profits relative to other occupations. Conversely, the share of unemployed individuals declines due to improved labor market conditions. The self-employed share rises on impact as higher productivity temporarily raises the return to self-employment, but it quickly declines as higher wages increase the return to wage work and induce some self-employed entrepreneurs to transition into employment.



(a) TFP Shock

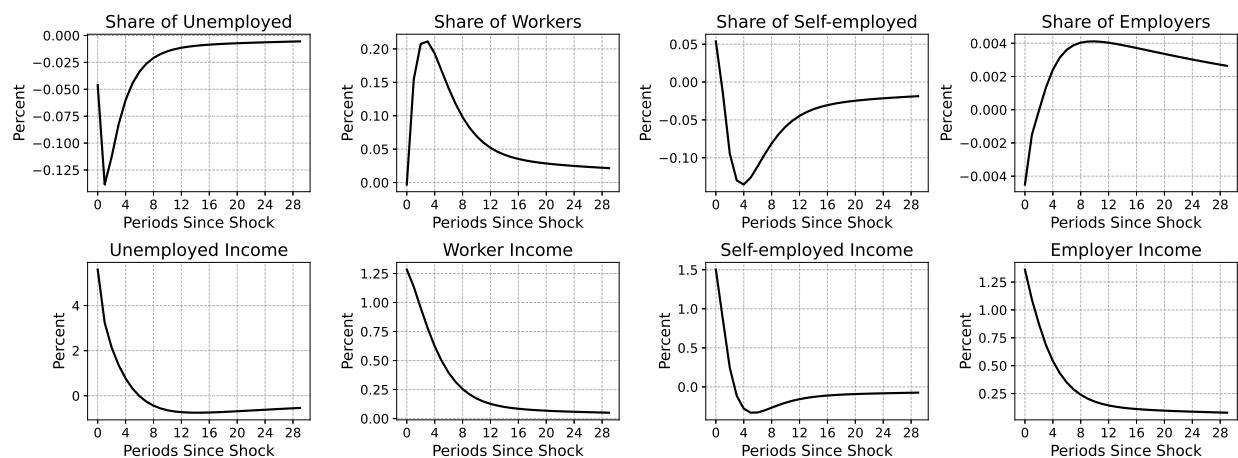


(b) Financial Shock

Figure 15: Impulse Response Analysis - Aggregate Variables

Note: These figures show the impulse response of aggregate variables and prices to a one percent positive shock. All variables except interest rate are in percent deviations from the steady state.

Turning to occupational-level income, unemployed individuals experience the largest percentage increase, although this mainly reflects their very low baseline income. On



(a) TFP Shock



(b) Financial Shock

Figure 16: Impulse Response Analysis - Population Dynamics

Note: These figures show the impulse response of population shares and group-level incomes to a one percent positive shock. All variables are in percent deviations from the steady state except for population shares, which are in percentage point deviations.

impact, the income response is largest for the unemployed, followed by self-employed and employers, and finally workers.

The response to a positive financial shock is more concentrated among employer entrepreneurs. The population share of employers rises, while the shares of the other occupations decline. The shock raises aggregate output, but the effect is much smaller than in the TFP shock case because the financial shock primarily affects entrepreneurs, especially employers. Since wages decline slightly, worker income falls modestly. By contrast,

employer income rises as relaxed collateral constraints allow employer entrepreneurs to expand their scale of production.

E Sensitivity Analysis: Second Moments

Table 11: Sensitivity Analysis: Alternative Calibrations

	Baseline	Low κ	High κ	Low ψ	High ψ
Panel A: Steady-State Population Shares					
Unemployed	0.09	0.09	0.09	0.09	0.09
Workers	0.70	0.70	0.70	0.70	0.70
Self-employed	0.18	0.18	0.19	0.18	0.18
Employers	0.03	0.03	0.02	0.03	0.03
Panel B: Business Cycle Volatility (Relative Std. Dev. to Output)					
Consumption	0.54	0.54	0.53	0.50	0.63
Investment	2.95	2.95	2.96	3.10	2.61
Worker Share	0.24	0.24	0.24	0.09	0.58
Self-emp. Share	0.20	0.21	0.19	0.09	0.46
Employer Share	0.09	0.10	0.08	0.10	0.08
Self-emp. Income	1.04	1.05	0.99	0.98	1.98
Employer Income	4.87	4.76	5.26	5.08	4.43
Panel C: Correlation with GDP					
Consumption	0.79	0.80	0.79	0.76	0.86
Investment	0.94	0.94	0.94	0.95	0.93
Worker Share	0.74	0.74	0.75	0.56	0.82
Self-emp. Share	-0.65	-0.65	-0.64	-0.44	-0.77
Employer Share	0.70	0.68	0.72	0.68	0.74
Self-emp. Income	0.23	0.21	0.33	0.77	-0.47
Employer Income	0.82	0.82	0.82	0.81	0.84

Notes: This table reports key moments from the model under alternative calibrations. The “Baseline” column reports the moments from the main calibration. Each subsequent column changes one parameter relative to the baseline, holding all other parameters fixed. “Low κ ” and “High κ ” refer to a 90% decrease and a 500% increase in the fixed cost of employer entrepreneurship, respectively. “Low ψ ” and “High ψ ” refer to a 50% decrease and a 100% increase in the labor market cyclical parameter, respectively. Panel A reports steady-state population shares, Panel B reports standard deviations relative to aggregate output, and Panel C reports correlations with GDP.

Table 11 presents a sensitivity analysis of the model’s second-moment properties. The

baseline column reports the main calibration, while the remaining columns vary either the fixed cost of employer entrepreneurship, κ , or the labor market cyclical parameter, ψ . The purpose of this exercise is to assess whether the model's business cycle implications depend on a narrow calibration of these two parameters.

Changing the fixed cost parameter mainly affects the composition of entrepreneurs, as expected. Lowering κ has little effect at the reported precision because the fixed cost is already small in the baseline, and entry into employer entrepreneurship also requires sufficient wealth and managerial ability. Raising κ , by contrast, makes employer entrepreneurship less attractive: the employer share falls from 0.03 to 0.02, while the self-employed share rises from 0.18 to 0.19. The worker and unemployed shares are essentially unchanged. The second moments are also broadly stable across these fixed-cost variations. Employer income remains the most volatile income component, and the population share of self-employed entrepreneurs remains countercyclical.

The labor market cyclical parameter, ψ , has no effect on steady-state population shares because it governs how labor-market transition probabilities respond to deviations of output from steady state. Its main effect is therefore on business-cycle dynamics. When ψ is low, occupational transitions respond less to aggregate conditions, so the volatilities of worker and self-employed population shares fall substantially. When ψ is high, labor-market flows become much more sensitive to the business cycle. The relative standard deviation of the worker share rises from 0.24 in the baseline to 0.58, while the relative standard deviation of the self-employed share rises from 0.20 to 0.46.

The high- ψ calibration also strengthens the countercyclical role of self-employment. The correlation between the self-employed share and GDP becomes more negative, falling from -0.65 in the baseline to -0.77 . In this case, movements into self-employment during downturns are large enough that self-employed income becomes countercyclical. This explains why the correlation between self-employed income and GDP turns negative under the high- ψ calibration. By contrast, when labor-market cyclical parameter is weaker, self-employed income is more strongly procyclical because the extensive-margin response into self-employment is muted.

Overall, the sensitivity analysis shows that the main qualitative patterns are robust.

Employer income remains highly volatile and procyclical, the employer population share remains procyclical, and the self-employed share remains countercyclical across all calibrations. Varying κ changes the steady-state composition of entrepreneurs in the expected direction, while varying ψ mainly changes the strength of occupational movements over the cycle.